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Multi-Person Human Mesh Recovery with the Anny Body Model

Master's Thesis

by

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Eigenständigkeitserklärung

Hiermit versichere ich, dass ich die von mir vorgelegte Arbeit mit dem Thema „Multi-Person Human Mesh Recovery with the Anny Body Model“ selbstständig verfasst habe, dass ich die verwendeten Quellen und Hilfsmittel vollständig angegeben habe und dass ich die Stellen der Arbeit — einschließlich Tabellen und Abbildungen —, die anderen Werken oder dem Internet im Wortlaut oder dem Sinn nach entnommen sind unter Angabe der Quelle als Entlehnung kenntlich gemacht habe.

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Muhammad Saad Najib

Use of AI tools

An AI coding assistant (Claude, Anthropic) was used in this work under my direction: for writing and debugging code, for the analysis scripts that read the training logs and produce the tables and plots, and for drafting and revising the text of this thesis. Every number reported here was checked by me against the logs, scripts and checkpoints named in Appendix A, and the repository history records the assistant's contributions. I take full responsibility for the content of this thesis and can explain every component of it. **[TODO: Student: confirm the wording against RPTU's current rules on AI disclosure and sign off.]**

Abstract

Multi-person human mesh recovery finds every person in one image and builds a 3D body mesh for each. This thesis moves Multi-HMR, a network that does this in one forward pass, from the SMPL-X body model to Anny, an open body model whose shape is set by named values such as age, gender and height. I train it on the synthetic AnnyOne images and on the real 3DPW images, whose SMPL ground truth I converted to Anny with my own fitting script.

The model I started from showed three visible failures: thin, twisted necks, claw-like fingers, and adult bodies on children. Checks of the labels, the body model and the gradients ruled out a rig or decoding problem. Instead, I found a depth bug: the network places each body by its head, but my training targets were built at the pelvis, so it learned to put people too far away and to draw them too small. I also found that two neck bones and two hand bones were turned by half a turn; a constant rotation at the output, with no new parameters, turns them back. In the converted 3DPW labels I found and fixed three defects (reversed gender, a missing actor in two-person frames, missing labels), fitted all 87 people again and checked the new labels independently.

The final model combines both fixes with 30 % real images in every batch. A control run has the same training without the two fixes. On 100 held-out AnnyOne images the final model has a mean vertex error (PVE) of 73.7 mm, against 77.4 mm for the control and 70.5 mm for the starting model. These gaps lie within the 7.6 mm that separated two identical training runs. The errors the fixes target moved far more: against the control, the error of the body's 3D position fell from 1,402 to 728 mm, the neck02 bone error from 175.8 to 4.8 degrees and the finger error from 17.4 to 8.3 degrees. On the 3DPW test set, measured against my converted labels, the final model reaches 89.0 mm PVE, against 126.9 mm for NAVER's released Anny model and 121.8 mm for the starting model; most of this gain comes from training on real images at all. On real photos with age and gender labels (Relative Human), gender accuracy rose from 46.5 to 64.2 % and the separation of children from adults (AUC) from 0.61 to 0.72. A ViT-B student distilled from the final model has 3.14 times fewer parameters and 3.19 times fewer FLOPs, but misses the target of 95 % of the final model's accuracy on both test sets (110.1 mm PVE on 3DPW); it is still better than NAVER's model there.

Three things remain open. Children are a measured improvement, not solved. Near and far people still receive similar depths: the fix moved the depth offset and the body size, not the depth slope (0.42, where 1 is correct). Every result comes from one training run per model, without a fixed seed.

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1. Introduction

1.1. The task

Human mesh recovery (HMR) estimates the 3D body surface of the people in an image. Methods predict the parameters of a parametric body model, a function that turns a pose and a few shape values into a closed mesh. Multi-HMR [Bar+24] does this for every person in an image in one forward pass: a Vision Transformer turns the image into patch tokens, a detection head finds one anchor point per person, and a cross-attention head outputs each person's pose, shape and distance from the camera. In this thesis the output is a body of Anny [Bré+26], a body model released in 2025: for every person the network returns the rotations of Anny's 163 bones, its 11 shape values and the body's place in 3D. I start from the public ViT-L checkpoint that NAVER released for Anny, keep its backbone, train new heads on synthetic images and then fine-tune on real ones (Figure 4.1).

When I began the work reported here, the model I had trained (Base) showed three failures that anyone could see in its renders. Necks were thin, twisted stalks under the head (Figure 5.1). Fingers curled into claws (Figure 6.4). A boy in one of our own photos received an adult body. The usual error measure, the mean vertex error after the pelvis is aligned (PVE), was 70.5 mm on held-out synthetic images and showed none of this. Behind the visible failures lay one that no render showed: the network placed people far behind their true position and drew them about a third too small. PVE could not see it, because it removes the body's position before it compares.

A large part of this thesis is therefore about finding out why the model failed before changing it. I tested each assumed cause on its own: a property of the rig for the neck, the image resolution for the fingers, an age that was never learned for the children. The rig and the age explanations were wrong, and resolution explains little of the finger error. The fixes that followed are small: training targets built at the right joint, and a constant rotation of four bones at the output. What made them possible were measurements that can see each failure.

1.2. Why Anny

Almost every HMR system of the last decade predicts a member of the SMPL family [Lop+15; RTB17; Pav+19]. SMPL is released under a research-only licence. Its shape space is learned from scans of adults, so its ten shape coefficients have no names, and children need a separate infant model [Hes+18; Sun+22]. Anny's code is under the Apache 2.0 licence and its assets under

Req.	Tier	What it asks for	Main evidence
1a-pose	1	pose of all body parts looks correct on the 3DPW test set and on any real photo	Sections 6.3–6.6
1a-shape	1	size, proportions, age, muscle and the other shape values visibly change with the person	Sections 6.7, 6.8
1b	1	not exactly Multi-HMR: new and better on both test sets, or much smaller with $\geq 95\,\%$ of the accuracy	Sections 6.9, 6.10
1c	1	evaluation with Anny parameters on a real data set	Chapter 3, Section 6.6
2a	2	3DPW train and test converted to Anny; the train set used in training	Chapter 3, Section 6.6
2b	2	the model uses motion in video instead of single frames	Chapter 8 (plan only)
3	3	a well-written report; a clean codebase and verifiable models	this thesis, Appendix A

Table 1.1.: *The supervisor's requirements in short form. Tier 1 is required for submission, tier 2 recommended, tier 3 for the best grade. Status words and the full evidence are in Table 7.1 (Section 7.1).*

CC0 [NAV26a]. Its shape is set by 11 named *phenotype* values, among them gender, age, muscle and weight, so a baby, a child and an adult come from the same model, and a supervisor can ask whether the body visibly changes with the person (Section 2.1).

Anny comes with AnnyOne [Bré+26], a large set of synthetic images with exact labels, and NAVER has released Multi-HMR-Anny, a Multi-HMR checkpoint trained on it. A multi-person HMR system for Anny therefore already exists; it is where my backbone comes from and the baseline I compare against. What did not exist were real images with Anny labels. I therefore converted the SMPL ground truth of 3DPW [Mar+18], a real video data set, to Anny.

1.3. What the thesis had to deliver

My supervisor set the goals of the thesis as requirements in three tiers (Table 1.1). Section 7.1 returns to the list and gives each requirement a status word, its evidence and what is missing. The outcome on 18 October 2026: 2a is met, 1c is met with caveats, 1a is partly met, 1b depends on a student model whose training was running, and 2b had not been started.

1.4. Contributions

Every run was trained once, without a fixed seed. Two identical runs differed by 7.6 mm PVE (Section 5.7), so I compare each change with a matched control. Final is the thesis model, Control the same training without the two fixes (Table 4.2).

1. **A diagnosis of three visible failures.** Six structural checks rule out

a rig or decoding problem, and each failure is traced to a specific mechanism (Chapter 5).

2. **The depth anchor and its fix.** The targets were built at the pelvis while the network places people by the head; fixing this cuts the root error from 1,402 to 728 mm against Control (Sections 5.3, 4.4, 6.2).

3. **The bone correction.** Two neck and two hand bones are half-turned in every Base checkpoint; a constant rotation at the output brings the neck02 error from 175.8 to 4.8 degrees (Sections 5.4, 4.5, 6.3).

4. **3DPW labels in Anny space.** A converter from SMPL to Anny; I found and fixed three defects in the first label set, and the median joint error fell from 18.7 to 14.1 mm (Section 3.3, Table 3.4).

5. **Final against a matched control.** On the 3DPW test set Final has the lowest error on our labels, 89.0 mm PVE against 91.5 for Control and 121.8 for Base (Sections 6.5, 6.6).

6. **Body shape on real photos.** On Relative Human, Final raises gender accuracy from 46.5 to 64.2 % and the child-against-adult AUC from 0.606 to 0.719; shape masks trade the gender gain for body variety (Sections 6.7, 6.8).

7. **The baseline, measured with the same code.** Multi-HMR-Anny reaches 126.9 mm PVE on 3DPW against 89.0 mm for Final (Section 6.9).

8. **A smaller student for requirement 1b.** A ViT-B Student distilled from Final has 3.14× fewer parameters and runs 2.10× faster, but misses the 95 % bar on both test sets (119.6 and 110.1 mm); it beats Multi-HMR-Anny on 3DPW (Section 6.10).

1.5. Outline

Chapter 2 gives the background. Chapter 3 describes the data and the 3DPW conversion. Chapter 4 gives the network, the losses, the two fixes and the evaluation protocol. Chapter 5 shows how each failure was isolated, and Chapter 6 reports the results. Chapter 7 goes through the requirements and the limitations; Chapter 8 concludes.

2. Background and Related Work

This chapter introduces the body models, networks, data and training ideas the thesis builds on, and for each prior work it says which choice of mine it explains or constrains. What the thesis adds is listed in Section 1.4.

2.1. Parametric body models: the SMPL family and Anny

A parametric body model is a differentiable function $M(\boldsymbol{\theta}, \boldsymbol{\beta}) \to \mathbf{V} \in \mathbb{R}^{N\times 3}$ that maps a pose $\boldsymbol{\theta}$ and a shape $\boldsymbol{\beta}$ to the N vertices of a mesh. The shape deforms a template mesh in its rest pose. The pose moves a skeleton of bones, and the skin follows by *linear blend skinning* (LBS):

$$\mathbf{v}'_i \;=\; \sum_k w_{ik}\, \mathbf{G}_k(\boldsymbol{\theta}, \boldsymbol{\beta})\, \mathbf{v}_i(\boldsymbol{\beta}), \tag{2.1}$$

where $\mathbf{v}_i(\boldsymbol{\beta})$ is the shaped rest vertex, w_{ik} the skinning weight of bone k and $\mathbf{G}_k$ the bone's world transform, found by chaining the rotations of all bones from the root down to k. Because the transforms chain, a wrong rotation high in the tree moves everything below it. This matters in Chapter 5, where two neck bones turned by half a turn leave the head pointing the right way but twist the skin between them.

The SMPL family. SMPL [Lop+15] has 6,890 vertices and 24 joints. Its shape space is a set of principal components learned from registered scans of adults; most methods use the first 10 coefficients. These are statistical directions, not named properties: no coefficient means "older" or "more muscular". Children lie outside the space. SMIL [Hes+18] is a separate infant model, and AGORA [Pat+21] and BEV [Sun+22] blend the adult template towards an infant one with one extra coefficient. SMPL-H [RTB17] adds MANO hands with 15 joints each, and SMPL-X [Pav+19] adds a face, for 55 joints on 10,475 vertices. SMPL is licensed for non-commercial research only, and every SMPL-based network and dataset inherits these terms. Multi-HMR, the network this thesis starts from, predicts SMPL-X [Bar+24], so moving it to Anny meant replacing the whole output layer.

Anny. Anny [Bré+26] is a body model by NAVER LABS Europe built from the open MakeHuman project [Mak] instead of from scans. Its code is under the Apache 2.0 licence and its assets under CC0 [NAV26a]. I use version 0.6.1 with the full rig of 163 bones and a mesh of 13,718 vertices, the rig of the



Figure 2.1.: *Anny at the mean AnnyOne shape in rest pose, with only the age value changed from 0 to 1. The skeletal height (head joint to mean ankle) grows from 0.53 m to 1.54 m. Body model only, no network; one camera for all panels, so the relative size is visible. The approximate years printed in the panel titles come from one of three age scales that disagree (Section 2.1) and are not used in this thesis.*

AnnyOne labels and of NAVER's checkpoint (the pruned 104-bone default rig of newer versions is not used). Pose is one rigid transform per bone, applied by Equation (2.1). Of the bones, 88 are helper bones (toes, breasts, face micro-bones) whose rotation an image cannot reveal; I keep them at rest, which leaves 75 predicted bones. The neck is modelled by three stacked bones, `neck01` to `neck03`, followed by `head`.

Shape is set by 11 *phenotype* values between 0 and 1: `gender`, `age`, `muscle`, `weight`, `height`, `proportions`, `cupsize`, `firmness`, and three blend weights `african`, `asian` and `caucasian`. Each value drives artist-made blend shapes, vertex offsets of the rest mesh. Because the values have names, a predicted body can be read as "older" or "heavier". Gender 0 is the male end and 1 the female end (rest stature 1.753 m and 1.632 m). Age blends five anchor bodies placed at linspace$(-\frac{1}{3}, 1, 5)$: age 0 is the baby, 0.333 the child, 0.667 the young adult and 1 the old adult; the newborn would need an age below 0, which a network output squashed into $(0, 1)$ never gives.

Figure 2.1 shows what age does. At the mean AnnyOne shape, moving age from 0 to 1 raises the skeletal height from 0.53 m to 1.54 m (1.37 m at 0.5); most of the change happens below about 0.5. A small height value does not make a child: at age 0.6 the smallest reachable body is 1.22 m tall and keeps adult proportions. What marks a child is narrow shoulders relative to the head, a shoulder-to-head width ratio of 2.24 at age 0.1 against 3.00 at 0.6. Only a low age produces it, which is why Chapter 5 treats children as an age problem. I never convert ages to years, because three scales are in use and disagree: the value read as 11 years is 0.415 in Anny's calibration, 0.333 in the code I inherited and 0.19 on the MakeHuman slider. I call a person *child-shaped* if the age value is below about 0.45.

AnnyOne. AnnyOne is NAVER's synthetic dataset for Anny: rendered scenes of several people with exact Anny labels (all bone transforms, all phenotypes and the camera). Its authors describe more than 780,000 images [Bra+26]; the release copy I train on holds 579,960 images. It is the only source of complete Anny labels, shape included, that I have (Section 3.1).

2.2. Single-person and multi-person human mesh recovery

Regression and fitting. HMR [Kan+18] regressed body-model parameters for one cropped person with a convolutional encoder, an iterative regressor and an adversarial pose prior. HMR 2.0 [Goe+23] replaced the encoder by a Vision Transformer (ViT) [Dos+21] and showed that a large ViT trained on a broad data mixture is a strong and simple recipe, which Multi-HMR shares. My model follows the same principle: the network outputs the parameters directly, with no optimisation at test time. SMPLify [Bog+16] goes the other way and fits SMPL to 2D keypoints by optimisation; SPIN [Kol+19] puts such a fit inside the training loop. My 3DPW converter (Chapter 3) is an optimisation of the SMPLify kind, but in 3D, between two body models, and run once offline.

Single-shot multi-person: ROMP and Multi-HMR. Two-stage methods run a detector and then a single-person network on each crop, so the cost grows with the number of people. ROMP [Sun+21] does both in one pass: it predicts a heatmap of body centres and reads body parameters at the detected centres. Multi-HMR [Bar+24], the architecture of this thesis, keeps this idea. A DINOv2 [ODM+23] ViT turns the image into one token per 14×14-pixel patch, a 48×48 grid at 672 pixels. Each token scores whether a person's primary keypoint, the head joint in the released code [NAV26b], lies in it. High-scoring tokens become the queries of the *Human Perception Head* (HPH), which attends to all image tokens and regresses the body parameters and the person's distance from the camera. That distance belongs to the head joint, and this anchor is at the centre of the depth defect of Chapter 5. Section 4.1.2 gives the details. The released SMPL-X checkpoints reach a PVE of 94.1 mm on the 3DPW test set with ViT-L in 74 ms and 94.0 mm with ViT-B in 43 ms on a V100 [NAV26b], which is why my smaller model is a ViT-B student. These are SMPL-X errors and cannot be compared with the Anny-space errors of this thesis.

BEV and Relative Human. BEV [Sun+22] extends ROMP with explicit depth reasoning and an age-aware body. Its authors collected Relative Human, real photos in which every person carries an age group (adult, teenager, kid, baby), a depth layer and keypoints. It is the only real data with age labels I could use, so I use it, for evaluation only, to test whether predicted ages and genders follow the people on real photos (Section 3.7). It has no 3D bodies.

Multi-HMR-Anny and Anny-Fit. In February 2026 NAVER released a Multi-HMR checkpoint that predicts Anny, trained on BEDLAM, AnnyOne and further real images [NAV26b]. A multi-person HMR network for Anny therefore already exists, and this thesis does not claim to be the first. This checkpoint, Multi-HMR-Anny, is where my backbone weights come from and the baseline I compare against (Section 6.9). On the same backbone its decoder has 8

cross-attention blocks and 76.57 M parameters, against 2 blocks and 14.01 M in mine. It was trained on AnnyOne, which includes the images I hold out, so its AnnyOne error is most likely an in-sample number. Anny-Fit [Bra+26], by the same group, refines such predictions per image by optimisation and adapts them to children at test time; my model stays a single forward pass. I did not use it; it is the natural comparison for future work on children.

2.3. Synthetic training data

AGORA [Pat+21] renders scanned people with SMPL-X fits into realistic scenes, and BEDLAM [Bla+23] showed that synthetic images alone can train competitive networks. AnnyOne does the same for Anny. Synthetic labels are exact and complete, but leave open how well a network transfers to real photos. In this thesis my shape predictions follow the labels on the synthetic holdout but collapse towards the training average on real photos (Chapter 5). This is why Final mixes 30 % real images into every batch, and why those images need Anny labels.

2.4. Real-data labels for a new body model

3DPW [Mar+18] is the standard real-world test set for mesh recovery: 60 video sequences with SMPL ground truth for one or two actors each. SMPL and Anny share neither skeleton nor shape space, so the labels must be re-expressed in Anny by fitting. The SMPL-X code transfers fits between SMPL-family models through a vertex correspondence [Pav+19], and the Anny package offers an inverter that fits Anny to a target mesh [NAV26a]; none turns a 3DPW label into an Anny label directly, so I wrote my own converter (Section 3.3). A fit to joint positions recovers only what joint positions determine; what the converted labels can and cannot test is listed in Section 3.6. SHAPY [Cho+22] supervises shape on real images with measurements and word attributes instead; Anny's named phenotypes would suit such labels, but I collected none.

2.5. Knowledge distillation

Knowledge distillation [HVD15] trains a small *student* to reproduce a large, trained *teacher*, originally by matching its class probabilities softened by a temperature. FitNets [Rom+15] let the student also match an intermediate representation of the teacher through a small learned regressor. DINOv2 [ODM+23] distilled its smaller ViTs from its largest one, so a DINOv2 ViT-B is a good starting point for a student. My Student combines both forms: a projector maps its 768-wide ViT-B tokens to the teacher's 1024-wide tokens, which lets the teacher's heads be copied unchanged (Section 4.7). It has 101.38 M parameters against 318.38 M, and the question is whether it keeps at least 95 % of Final's accuracy on both test sets.

2.6. Temporal models for video

My model works on single frames, and the temporal requirement (2b) has not been started. Most video methods refine per-frame features of an image network: HMMR [Kan+19] with a temporal convolution, VIBE [KAB20] with a recurrent network and a motion discriminator tied to SMPL, TCMR [Cho+21] with branches that predict the current frame from its neighbours to reduce jitter, and MPS-Net [Wei+22] and GLoT [She+23] with attention over frames. SmoothNet [Zen+22] smooths the output poses of any estimator without image features. PHALP [Raj+22] and 4DHumans [Goe+23] track people through video, the first step of any multi-person video method, and WHAM [Shi+24] places the motion in world coordinates.

The plan for 2b (Section 8.2) is a small refiner of the first kind on the frozen Final model: it reads the per-person HPH tokens over a window of frames, is trained on 3DPW-train tracks, and is judged by acceleration error and PVE against SmoothNet-style smoothing, the baseline it would have to beat.

3. Data

This chapter describes the images and labels behind every number in the thesis. Table 3.1 lists every split with its size and its use. I converted 3DPW's SMPL ground truth to Anny so that the model can train and be tested on real people in Anny space (requirements 1c and 2a). Most of the chapter is about the 3DPW conversion, because the converted labels decide what the real-image numbers can show. Two rules hold throughout: no test image is trained on, and numbers on the old 3DPW labels never appear next to numbers on the v23 labels.

split	size (images / people)	used for	chapters
AnnyOne training	579,860 images	training of every model; 70% of each mixed batch	4, 6
AnnyOne holdout	100 images	headline synthetic evaluation (V1); never trained on	5, 6
diagnostic manifest	40 + 15 entries	per-person diagnostics; excluded from every fine-tune	5
3DPW-Anny train, v23	5,882 / 7,584	30% of each mixed batch	4, 6
3DPW-Anny val., v23	2,622 / 3,311	300 images scored after each epoch; monitoring only	6
3DPW-Anny test, v23	8,190 / 11,854	reported 3DPW numbers; never trained on	6
3DPW-Anny old labels	16,631 / 20,028	training of the history models only; retired	4
Relative Human val.	635 / 2,130	children and gender on real photos; evaluation only	5, 6
demo images	12 images	by-eye comparison; no metric	5, 6

Table 3.1.: *Every split of the thesis, its size and its use. 3DPW-Anny sizes are images / labelled people after the drop rules (v23 labels, Section 3.5); numbers on 3DPW are always against these converted labels, never against the original SMPL meshes. The final model is the last epoch of its run, so the 3DPW validation split does not select it.*

3.1. AnnyOne

AnnyOne [Bré+26] is the synthetic dataset released with Anny. It holds 579,960 rendered images of 1280×1280 pixels, with 2.73 people per image on average. For every visible person the label gives the exact Anny pose (163 bone transforms), the 11 phenotypes and the camera intrinsics. The network sees each image resized to 672 pixels. The field of view changes from image to

image (51° to 110° on a sample of five images), so training and evaluation use each image's own intrinsics.

Split. Indices 0 to 579,859 are the training set. The last 100 images, indices 579,860 to 579,959, are the holdout, my own choice and not an official split; no model trained in this thesis has seen them. NAVER's model was trained on the AnnyOne release, so for that model the holdout is most likely in-sample. The 3DPW test script also scores the last 100 AnnyOne images, but in setting V0; its "AnnyOne test-100" value is therefore not the holdout number, which is always given in setting V1.

Few children. In Anny the child shape appears below an age of about 0.4 to 0.5 (Section 2.1). Only 2.53% of AnnyOne's people are younger than 0.3 (1.06% below 0.2, 4.45% below 0.4); the median age is 0.601, and 41% of all people lie between 0.5 and 0.6. In a training batch, 2.1% to 2.3% of the supervised people are younger than 0.3. The holdout has nobody younger than 0.513 and tests adults only, which is why I added Relative Human (Section 3.7).

3.2. 3DPW

3DPW [Mar+18] is a real video dataset of 60 sequences filmed with a hand-held phone camera at 1080×1920 pixels (portrait) and 30 frames per second. Inertial sensors on the actors give SMPL ground truth. The standard release has 24 training, 12 validation and 24 test sequences, with one or two labelled actors each: 87 person-sequences (62 men, 25 women). Bystanders have no labels. I keep the splits as 3DPW defines them. One validation sequence, `outdoors_crosscountry_00`, yields no labels (Section 3.5), which leaves 11 usable validation sequences.

3.3. Converting SMPL to Anny

The converter (`smpl_to_anny.py`, version 2.3) fits an Anny body to every labelled person, frame by frame; a label builder moves the fits into 3DPW's camera frame and merges them into one label file per image.

Objective. Anny has no analytic inverse of its skinning, so I fit by gradient descent. For each person and frame I search Anny's 163 rotations $\mathbf{r}$ (6D), the phenotype logits $\boldsymbol{\ell}$ and a translation $\mathbf{t}$ so that Anny's joints land on the 24 SMPL joints $\hat{\mathbf{J}}^S$ of the frame:

$$\begin{aligned}\min_{\mathbf{r},\boldsymbol{\ell},\mathbf{t}} \; & \frac{1}{24}\sum_{k=1}^{24}\left\|\mathbf{p}_{\pi(k)}(\mathbf{r},\sigma(\boldsymbol{\ell}))+\mathbf{t}-\hat{\mathbf{J}}^S_k\right\|_2^2 \\ & +\lambda_r\sum_{b\neq\text{root}}\|\mathbf{R}_b-\mathbf{I}\|_F^2+\lambda_\phi\sum_{j\in\text{free}}\left(\sigma(\ell_j)-\tfrac{1}{2}\right)^2,\end{aligned} \tag{3.1}$$

where π is the joint correspondence and $\mathbf{p}_b$ the point attached to bone b: its head, or a virtual joint (Section 3.4). The first regulariser ($\lambda_r = 10^{-4}$) pulls every non-root rotation towards identity, because a twist about a bone's own axis moves no joint but corkscrews the mesh. The second ($\lambda_\phi = 10^{-3}$) pulls the free phenotypes towards 0.5, so that one remaining joint gap cannot drag a phenotype to an extreme. Adam (learning rate 0.05) fits the root, then the bone rotations, then the free phenotypes; shape comes last because it is the weakest signal. Each frame starts from the previous frame's solution (150 / 400 / 250 iterations per stage on a person's first frame, 20 / 80 / 60 afterwards). The targets are 3DPW's own per-frame `jointPositions`; the neutral SMPL model would move them by 37 mm on average, because 3DPW's shape coefficients belong to the gendered models.

Bones and shape. Only the root and 18 of the 23 mapped bones may rotate; 144 of 163 bones stay at rest, among them `neck02`, `neck03`, `head` and every finger bone. Joints cannot determine most phenotypes: weight, muscle, cup size, firmness and the three ethnicity values move no joint, and adult age barely does. The converter therefore reads gender from 3DPW, fixes age at 0.55 and the seven others at 0.5, and fits only height and proportions. Since version 2.3 it writes men as 0 and women as 1, Anny's convention (`-gender_convention anny`). After a first pass over all frames, the median of each free phenotype becomes the person's one body, and every frame's pose is fitted again with it (100 iterations).

Frames, camera and label files. I fit every third valid frame (10 frames per second). Since version 2.3 one grid holds for every person of a sequence (frame numbers divisible by 3, `-frame_grid shared`), so both actors are fitted on the same images. The builder writes one file per image in the AnnyOne format and drops a person when the fit error exceeds 80 mm, less than 30% of the mesh is inside the image, or the box is narrower or shorter than 40 px. A dataset class pads the portrait images to a square and shifts the principal point, so the training and evaluation code run on these labels unchanged.

3.4. Development of the converter

The first converter (v1) chose the Anny bones by name. It gave 34.5 mm on `downtown_bar_00`; on four test scenes the error was 30 to 37 mm on every frame, and all four actors came out with the same body. A residual independent of the pose and a shape independent of the person mean that the shape absorbed a fixed mismatch between the skeletons. Placing both models in their rest poses showed two wrong choices that no name reveals. Anny's rig numbers the spine top-down, SMPL bottom-up, so SMPL's lumbar `spine1` had been mapped to a bone 25 cm above it; and SMPL's hips had been mapped to `pelvis.L/R`, 4 cm from the hip joint, instead of `upperleg01.L/R`. Version 2 picks the spine bones by height along the body axis and the hip bones by

distance from the pelvis (Appendix B). Limbs and head fell to 1 to 10 mm, but the torso stayed at 43 to 73 mm (Table 3.2).

The reason is that the rigs define their torso joints at different anatomical points: SMPL's pelvis joint sits 9 cm above its hips, Anny's root at hip height and 7 cm in front. For nine torso joints I therefore measure once, in the rest pose, the offset $\boldsymbol{\delta}_b$ from the Anny bone head to the SMPL joint in the bone's own frame, and the fit moves the *virtual joint* $\mathbf{p}_b = \mathbf{h}_b + \mathbf{R}_b\boldsymbol{\delta}_b$ onto the SMPL joint, with $\mathbf{h}_b$ and $\mathbf{R}_b$ the posed bone head and frame. The offsets are 11.6 cm (pelvis), 4.3 cm (hips), 3 to 7 cm (spine) and 5 cm (collars). With them (v2.1) the torso fell to 13 to 22 mm and four varied test scenes to 15.8 to 20.1 mm (one occluded frame 43.5 mm). Age, however, came out at exactly 0.67 for every actor in every version: joints cannot tell one adult age from another. Version 2.2 fixes age at 0.55; on all 87 person-sequences it fitted 22,915 frames with a median error of 18.2 mm. By every measure the converter prints, these fits looked clean. They still held two of the three defects of the next section.

converter	mean (mm)	max	worst joints (mm)
v1: correspondence by name, neutral SMPL targets	34.3	37	all joints, common floor
v2: correspondence by geometry, 3DPW joints, gender from metadata (reversed until v2.3)	24.6	30	spine1 73, collars 55–59, spine3 58, pelvis 53
v2.1: + virtual torso joints, bone mask, warm start	15.8	18.8	shoulder 36, wrists 25–27

Table 3.2.: *Mean joint error over all 24 joints on* `outdoors_fencing_01` *(38 frames, including full lunges) for three converter versions. The geometric correspondence is worth about 10 mm, the virtual joints a further 9 mm.*

3.5. Three label defects and the v2.3 re-fit

The converter grades its own fit, so it cannot see errors in what it is asked to fit or in what happens to the fits afterwards. I therefore wrote an independent check that reads the finished label files. On sampled images from every sequence it compares the labels with 3DPW's joints, surface and 2D keypoints, confirms that the loader passes them on unchanged, and tests gender, a label for every valid actor and the head turn. On the first label set ("old labels", v2.2 fits, about 2,000 images) its grade was FAIL, with three defects (Table 3.4).

Defect 1: gender reversed. The converter wrote 1.0 for men and 0.0 for women, MakeHuman's convention and the reverse of Anny's, so every 3DPW man was labelled with a female body. Refitting with the flipped value lowered the joint error for 72 of 73 men (median 19.2 to 15.3 mm); for women it raised the joint error (13.7 to 16.3 mm) but brought the surface closer for 27 of 31.

	old labels	rebuilt	v23 labels
converter fits	v2.2	v2.2	v2.3
image files of 8 sequences	missing	re-extracted	re-extracted
train, images / people	4,451 / 5,183	6,437 / 7,589	5,882 / 7,584
validation	2,458 / 2,995	2,777 / 3,314	2,622 / 3,311
test	9,722 / 11,850	9,722 / 11,850	8,190 / 11,854
total	16,631 / 20,028	18,936 / 22,753	16,694 / 22,749
gender reversed	yes	yes	no
actor missing in two-person frames	yes	yes	no
labels missing (no image file)	yes	no	no
used for training	history models only	never	Final, Control, Student

Table 3.3.: *The three 3DPW-Anny label sets. All are built at every third frame with the same drop rules. The v23 labels have fewer images than the rebuilt set but the same people, because both actors of a two-person sequence now share each frame.* `outdoors_crosscountry_00` *yields no labels in any set (Section 3.5). Numbers measured on the old labels are retired and never appear next to numbers on the v23 labels.*

Defect 2: missing actor. In 407 of 960 sampled two-person images, one valid actor had no label. Each person's every-third-frame grid started at that person's own first valid frame, so after an invalid gap the two actors sat on different phases. The detection loss then treats a visible, real person as background.

Defect 3: missing labels. The training split had 4,451 label images where the fits implied 6,437 (31% missing). The image files of 8 sequences were incomplete on the cluster (`courtyard_captureSelfies_00`: 480 of 750 files), and the builder skipped frames without an image file without counting them. A rebuild after re-extracting the images recovered every label (Table 3.3, "rebuilt"), but still had defects 1 and 2, so I never trained on it.

The re-fit. Version 2.3 adds the two switches named above; their defaults reproduce v2.2 exactly. I re-fitted all 87 people with both switches on, and the post-check passed on all 60 sequences (Table 3.4, middle block). 9 single-person outdoor sequences became 1 to 3 mm worse, likely from a different warm start, which I did not test.

The v23 labels. I call the labels built from the v2.3 fits the v23 labels: 16,694 images and 22,749 people (Table 3.3). They have fewer images than the rebuilt set but the same people, because both actors now share each frame. No person was dropped for fit error or box size. 163 frames were dropped for visibility, all from `outdoors_crosscountry_00`: with 3DPW's own camera, its ground-truth joints sit at a depth of -0.3 m, behind the camera. The 3D fit is fine; image labels for this sequence cannot exist with this camera, and no label set ever contained it.

check	old labels	v23 labels
overall grade	FAIL	WARN
body joints vs 3DPW, median / p95 (mm)	18.7 / 24.5	14.1 / 21.1
surface, Anny→SMPL / SMPL→Anny (mm)	18.9 / 18.3	18.9 / 16.1
2D error over raw 3DPW (px)	+1.1	+0.4
gender labels	reversed	men 0, women 1
refit with gender flipped	better for 72 of 73 men	worse: men 13.9 → 20.0 mm
two-person images, an actor unlabelled	407 of 960	0 of 1,080
fitted frames; median frame error (mm)	22,915; 18.2	22,912; 14.0
head turn vs chest, labels / SMPL (°)	9.5 / 20.2	19.6 / 19.3
post-check of the re-fit, all fitted frames, 87 people		
mean joint error, men / women (mm)	19.0 / 15.0	14.7 / 15.2
two-person frames, a valid actor unfitted	4,457 of 8,274	0 of 6,055
still WARN on the v23 labels		
limb ratio Anny / SMPL, forearm / thigh		0.948 / 1.064
people with a joint more than 100 mm off		3.2%

Table 3.4.: *Independent conversion check of the 3DPW-Anny labels before (old labels, v2.2 fits) and after the re-fit (v23 labels, v2.3 fits). The check samples images from every sequence and compares the labels with 3DPW's own joints, surface and 2D keypoints. The middle block is the re-fit's own post-check over all fitted frames; old and new fits fitted different frames, so it is not a paired comparison.*

The check on the v23 labels. The same check on the v23 labels (3,314 people in 2,360 images, 59 sequences) moved the grade from FAIL to WARN (Table 3.4). The median body-joint error fell from 18.7 to 14.1 mm, flipping the gender now makes the fit worse, and no sampled two-person image misses an actor. The head turn relative to the chest is now kept; in the old test labels it was mostly lost (6.3° against 20.0° in SMPL). Two rows stay at WARN, the limb-length ratios and the people with a joint more than 100 mm off, mostly at the hip, likely because Anny's and SMPL's limb proportions and hip joints differ (Figure 3.1).

Figure pending: `F14_label_check.jpg`, three panels from the conversion check of the v23 labels, one of them a two-person frame.

Figure 3.1.: *Conversion check on the v23 labels: the 3DPW photo, 3DPW's SMPL ground truth with its 2D keypoints, and our Anny label drawn with 3DPW's camera. Real 3DPW frames. By design the labels have rest-pose hands and a head that turns only through* `neck01` *(Section 3.6).*

3.6. What the 3DPW labels can and cannot test

The v23 labels are converted labels, not ground truth. Four limits follow from the converter's design.

Eight constant phenotypes. The labels test three phenotypes: gender (from metadata), height and proportions (fitted; medians 0.47 and 0.51). The other 8 are the same for every person: age 0.55 and seven values at 0.5. A model trained on 3DPW is pulled towards these constants for every real person, by the shape loss and the mesh terms; 3DPW people are about 16% of the supervised people in a 70/30 batch (my estimate). The 3DPW error is also measured against these neutral-body meshes, so it likely rewards a neutral prediction; I did not test this. Section 4.6 describes the shape mask that removes the shape loss on the constants (Final-masked). At age 0.55 every label is a full adult (Figure 2.1); 3DPW has no children, so it can neither teach nor test them.

Head and hands at rest. `neck02`, `neck03` and `head` stay at identity in every label, so the head turns only through `neck01`, and the labels cannot test the half-turned neck bones of Chapter 5. 3DPW's SMPL ground truth has no finger joints, so every label has rest-pose hands; the training masks their rotation terms for 3DPW people (Section 4.6).

A floor of about 14 mm. The labels sit 14.1 mm (median) and 21.1 mm (p95) from 3DPW's own joints. Every Anny-space error on 3DPW contains this residual.

Not comparable with published numbers. I never evaluated against the original SMPL meshes, and our MPJPE averages all 163 Anny joints (Section 4.8), so I write "lowest error on our 3DPW-Anny labels". Bystanders are unlabelled, so precision on 3DPW is a lower bound.

3.7. Relative Human

Neither the holdout nor 3DPW contains children. Relative Human, introduced with BEV [Sun+22], fills this gap. It holds 8,741 real photos, many crowded and with children. Each person has a box, an age group (adult, teen, kid, baby) and a gender coded 0 for male, as in Anny; there are no 3D labels. The dataset ships without a licence file, so I use only the validation split and only for evaluation: 635 images with 2,130 labelled people (1,680 adults, 101 teens, 209 kids, 117 babies, 23 of unknown age).

The model runs once on each whole photo with a field of view of 55°, and detections are paired with labelled boxes as in Section 4.8. A predicted age below 0.45 counts as child-shaped, a predicted gender below 0.5 as male. I report recall and mean predicted age per age group, the child-against-adult AUC of the predicted age (kids and babies against adults) and the gender

accuracy. Every number covers detected people only, so recall always stands next to it.

3.8. The diagnostic manifest and the demo images

The checks of Chapter 5 need the same people every time. A seeded script picks them from the AnnyOne labels alone and stores 55 fixed (image, person) entries: 40 from the training range (adults, the youngest children, clearly visible hands, the hardest head turns) and 15 from the holdout. Base saw the training entries; every later fine-tune gives their images a sampling weight of zero, so they remain the same in-sample reference for all models. The manifest children therefore show whether a model can fit a child it has seen, not whether it generalises. For by-eye comparisons I use 12 real images: 4 photos and 8 3DPW test frames, one middle frame from each of 8 test sequences. No model trained on them and no metric is computed on them.

4. Method

This chapter describes the network, its losses, the training runs, the model names (Section 4.3) and the evaluation. The two changes that matter most are in Sections 4.4 and 4.5: the training targets now describe the joint the network anchors, and a constant rotation at the output turns four flipped bones back. Chapter 5 shows how the defects were found; Chapter 6 measures what the changes did.

4.1. Multi-HMR for Anny: task and architecture

4.1.1. The task

The input is one RGB image with camera intrinsics $\mathbf{K}$ (from an assumed field of view when they are unknown). It shows an unknown number of people. For each person i the network predicts the parent-relative rotations $\mathbf{R}_i \in SO(3)^{163}$ of Anny's bones, the phenotypes $\boldsymbol{\phi}_i \in [0,1]^{11}$, and the image position (u_i, v_i) and depth z_i of one anchor joint a, the head. The Anny body model M [Bré+26] turns $(\mathbf{R}_i, \boldsymbol{\phi}_i)$ into vertices and joints, and the mesh is moved so that its anchor joint sits at the back-projected anchor pixel:

$$\mathbf{V}_i = M_V(\mathbf{R}_i, \boldsymbol{\phi}_i) - M_{J,a}(\mathbf{R}_i, \boldsymbol{\phi}_i) + \mathbf{t}_i, \qquad \mathbf{t}_i = z_i\, \mathbf{K}^{-1}\,(u_i, v_i, 1)^\top. \tag{4.1}$$

The goal is a mesh with the right pose, shape and place in the camera frame, from one pass over the whole image.

4.1.2. Architecture

The network is Multi-HMR [Bar+24] with its output resized for Anny (Figure 4.1). I did not change the backbone, the detection or the cross-attention head, so the architecture is no answer to requirement 1b; the route to 1b is the smaller student of Section 4.7.

A DINOv2 ViT-L [ODM+23] with weights from NAVER's Anny checkpoint turns the image into tokens; one MLP scores each token for an anchor joint (the head, `--person_center head`) and a second predicts its 2D offset. Each detected token becomes a query of the HPH, which lets the people attend to each other and to all image tokens. The Anny head is what differs from the SMPL-X version: the rotations use the 6D representation [Zho+19] with Gram–Schmidt, the phenotypes a sigmoid, and the depth Multi-HMR's focal-normalised log-depth. Right after Gram–Schmidt four bones are multiplied by one constant rotation each (Section 4.5); then a fixed mask sets 88 helper bones (toes, breasts, face, subdivision bones) to the identity, in the prediction

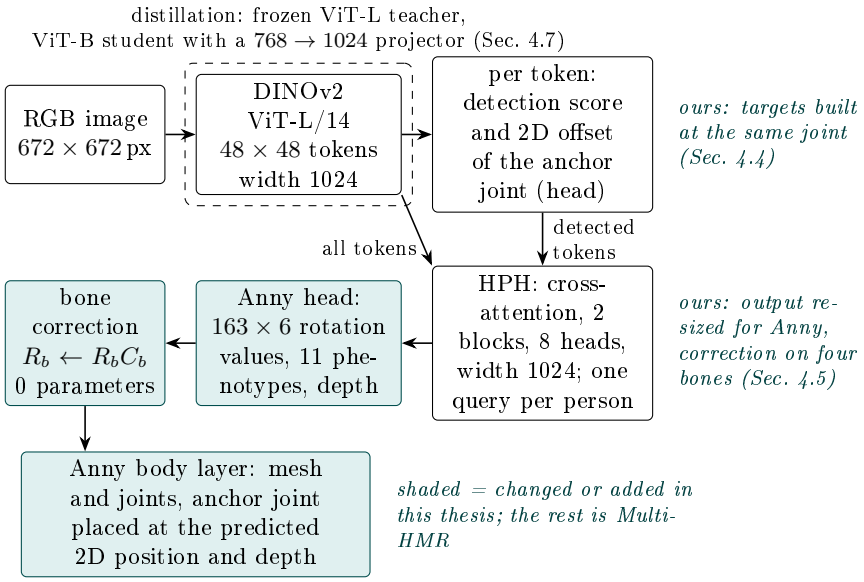


Figure 4.1.: *The network. A 672-pixel image passes through a DINOv2 ViT-L backbone (48×48 tokens). Every token gets a detection score and a 2D offset for the anchor joint, the head. For each detected person a cross-attention head (HPH) predicts 163 × 6 rotation values, 11 phenotypes and a depth; a constant correction turns four flipped bones back; the Anny layer builds the body and places it at the predicted distance. Shaded boxes are changed or added in this thesis; the dashed box marks the part that distillation replaces.*

and the ground truth alike, and keeps 75. The Anny layer runs inside the network and places the body with Equation (4.1), so the 3D and 2D losses reach the network through it. The network has 318.38 M parameters (heads 14.01 M) and needs 1,927.4 GFLOPs per image with one person.

4.2. Losses in plain words

Anny runs on each person's true parameters to give target vertices and joints; the anchor targets are built from one joint (Section 4.4). The total loss is

$$\begin{aligned}\mathcal{L} = {} & \alpha_{\text{det}}\mathcal{L}_{\text{det}} + \alpha_{\text{off}}\mathcal{L}_{\text{off}} + \alpha_{\text{rot}}\mathcal{L}_{\text{rot}} + \alpha_{\text{shape}}\mathcal{L}_{\text{shape}} \\ & + \alpha_{\text{dist}}\mathcal{L}_{\text{dist}} + \alpha_{\text{t}}\mathcal{L}_{\text{t}} + \alpha_{\text{j3d}}\mathcal{L}_{\text{j3d}} + \alpha_{\text{v3d}}\mathcal{L}_{\text{v3d}} \\ & + \mathbb{1}[e \geq e_{\text{2D}}]\,(\alpha_{\text{j2d}}\mathcal{L}_{\text{j2d}} + \alpha_{\text{v2d}}\mathcal{L}_{\text{v2d}}) + \lambda_{\text{kd}}\mathcal{L}_{\text{kd}} + \lambda_{\text{kd-out}}\mathcal{L}_{\text{kd-out}},\end{aligned} \tag{4.2}$$

with e the epoch; the distillation terms exist only for a student. Table 4.1 says what each term compares and gives its weights.

The detection term is the focal loss of CornerNet [LD18] (the log calls it `bce`, but it is not a binary cross-entropy). The neck and hand boost moves rotation loss between bones, it does not add any. The shape L1 is smallest at the median of the plausible answers, so weak image cues pull predictions toward the training median (Chapter 5). The 3D terms align both skeletons

Term (log name)	What it compares, in plain words	Default	Final	Control	Student
Detection (`bce`)	focal loss on the per-token score "an anchor lies here"	10	10	10	10
Offset (`offset`)	L1 on the anchor's position inside its token	1	1	1	1
Rotations (`rotmat`)	L1 over the 9 entries of each parent-relative rotation	0.1	0.1	0.1	0.1
neck and hand boost	renormalised: $\times 2.41$ neck, head, hands; $\times 0.60$ rest	1	4	4	4
Shape (`shape`)	L1 on the 11 phenotypes after the sigmoid	1	20	20	20
Distance (`dist`)	L1 on the anchor's focal-normalised log-depth	1	5	1	5
Translation (`transl`)	L1 on the anchor's 3D position in the camera frame	1	5	1	5
3D joints (`j3d`)	L1 on the 163 joints, each at its own pelvis	100	100	100	100
3D vertices (`v3d`)	the same for the mesh vertices	100	100	100	100
2D joints (`j2d`)	L1 in pixels on the joints projected from anchor and depth	1	0.1	0.1	0.1
2D vertices (`v2d`)	the same for the projected vertices	1	0.1	0.1	0.1
first 2D epoch	`--start_2d_epoch`	10	0	0	0
Global rot. (`global_rot`)	L1 on camera-frame head, neck03, wrist orientation	0	0	0	0
Feature KD (`kd`)	KL of softened student and teacher tokens, $T = 4$	–	–	–	1.0
Output KD (`kd_out`)	L1 to the teacher's rotations, phenotypes, depth	–	–	–	1.0
Target joint	`--gt_center_joint` (four anchor targets)	pelvis	head	pelvis	head
Bone correction	neck02, head, metacarpal1.L, finger4-1.R	none	on	none	on

Table 4.1.: *Loss terms and their weights. "Default" is the value in `loss.py`; Final, Control and Student are the values of those runs (model names in Table 4.2). The total loss is the weighted sum of all rows above the rule. For people from 3DPW the rotation and global-rotation terms skip the 38 finger, thumb and metacarpal bones in all three runs; the mesh terms are not masked. The shape mask for 3DPW people is off in all three runs and on only in Final-masked (Section 4.6). The Student keeps the teacher's bone correction during training and is evaluated with its own, re-estimated after training (Section 4.7.2).*

at the pelvis, so only distance and translation see placement directly, and the 2D terms indirectly; before the depth fix the two direct terms made up about 2.5% of the loss, against 25.7% for shape. Training runs in fp32 with TF32 products (mixed precision gave NaN values with the Anny layer), and steps with NaN or Inf gradients are skipped (16 of 10,000 in Final). No weight was tuned in this thesis.

4.3. Training schedule and model names

The Base lineage. Training began from NAVER's public ViT-L checkpoint, Multi-HMR-Anny. Its head has a different structure, so only the backbone was transferred. Four stages, L-stage 1 to 4, and a fifth run with shape weight 20 led to *Base* (epoch 99): synthetic images only, and the start of every fine-tune here. Continuations used a learning rate ten times below the cold start, which on a partly trained model had caused hundreds of NaN gradient events per epoch.

Final and Control. Final starts from Base and trains 10 epochs of 1,000 iterations at batch 4 at a constant learning rate of 10^{-5}; the last 4 backbone blocks and the heads train. Three changes come on together: the depth fix with distance and translation weight 5 (Section 4.4), the bone correction (Section 4.5), and 30% real images per batch with the hand mask (Section 4.6). The 55 images of the diagnostic manifest (Chapter 3) are never sampled. Training took 4,292 s on one A100. Control is identical but keeps the pelvis targets with weight 1 and has no bone correction, so Control against Final isolates the two fixes, and Base against Control shows the real images plus the restart cost. Each fix was tested alone only in short pilots (Chapter 5).

Model names. Table 4.2 is the key to the names used from here on; each name is one run. Table A.1 in Appendix A gives each name's run, checkpoint, starting weights, data, schedule and flags.

Name	Meaning
Multi-HMR-Anny	NAVER's released ViT-L model for Anny; source of every backbone, baseline of requirement 1b
L-stage 1 to 4	the transfer steps from Multi-HMR-Anny's backbone to Base
Base	end of the schedule, AnnyOne only; the start of every fine-tune
Mixed-L	Base fine-tuned with 30% 3DPW on the old labels; the previous best on 3DPW
Fresh-heads-L	fresh heads on Multi-HMR-Anny's backbone, 30% 3DPW (old labels) from the start
Control	Base, 10 epochs, 30% 3DPW (corrected labels), hand mask, neither fix
Final	Control plus the depth fix and the bone correction; the thesis model and the teacher
Final-masked	Final plus the shape mask on all 8 converter constants
Final-partial-mask	Final plus the shape mask on the 7 constants other than age
Student-S/1 to /final	the ViT-S students of Base, before both fixes; history
Student	the ViT-B student distilled from Final; the answer to requirement 1b

Table 4.2.: *Model names used in this thesis. Every name is one run; no run was repeated with another seed. Full definitions: Table A.1 (Appendix A).*

4.4. The depth anchor and its fix

The mismatch. The network places each person by the head: the detection token, the offset and the predicted depth describe the head, and the mesh hangs from it (Equation (4.1)). For AnnyOne images, however, the training code built the translation, distance, token and offset targets at joint 0, the pelvis. Every checkpoint before this thesis, Base included, was trained on these targets, so prediction and target described two joints about half a body apart. A mesh hanging from a head placed at the pelvis is drawn too low; the 2D terms can shrink this misfit by moving the body away, where it is drawn smaller, and neither the pelvis-aligned 3D terms nor the weak direct depth terms object (Figure 4.2). The mismatch is certain; the mechanism is likely but was not isolated. Section 5.3 gives the evidence: all 222 holdout people placed too far, drawn at 0.63 of their size.

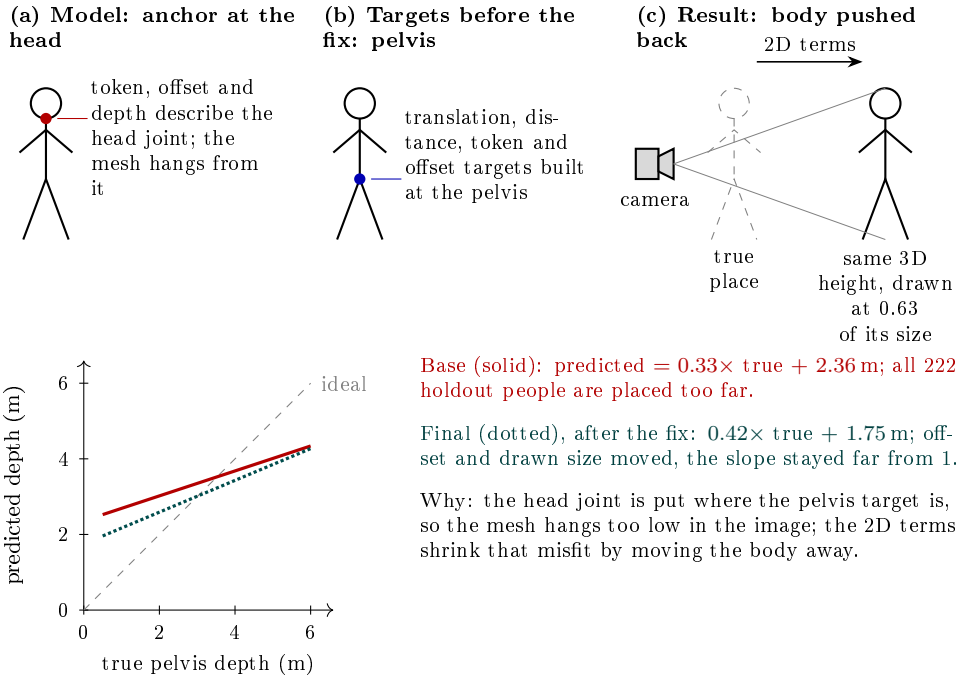


Figure 4.2.: *The depth-anchor bug. (a) The model places each body by its head; (b) the training targets were built at the pelvis; (c) the 2D terms reconcile the two by moving the body away, where the same 3D body is drawn smaller. Bottom: the fitted line of predicted against true pelvis depth for the 222 matched people of the AnnyOne holdout (setting V1), Base and Final. Schematic; the measured scatter is Figure 6.2.*

The fix. A single training switch (`--gt_center_joint person_center`) builds all four anchor targets at the joint the model anchors; the visibility filter and the one-person-per-token rule move with them. The network does not change. I also raised the distance and translation weights from 1 to 5, because they are the only direct depth terms; the pilots behind this choice are

in Table 5.3. Final uses the switch with weight 5; Control keeps the pelvis targets with weight 1.

4.5. The bone correction

What it corrects. A per-bone dump on the holdout (Section 5.4) showed neck02 and the head bone turned half a turn in all 8 checkpoints of the Base lineage. The half-turns cancel, so the head points the right way and PVE never showed the flip, but the neck skin between them is twisted to 0.46 of its true girth (Figure 4.3).

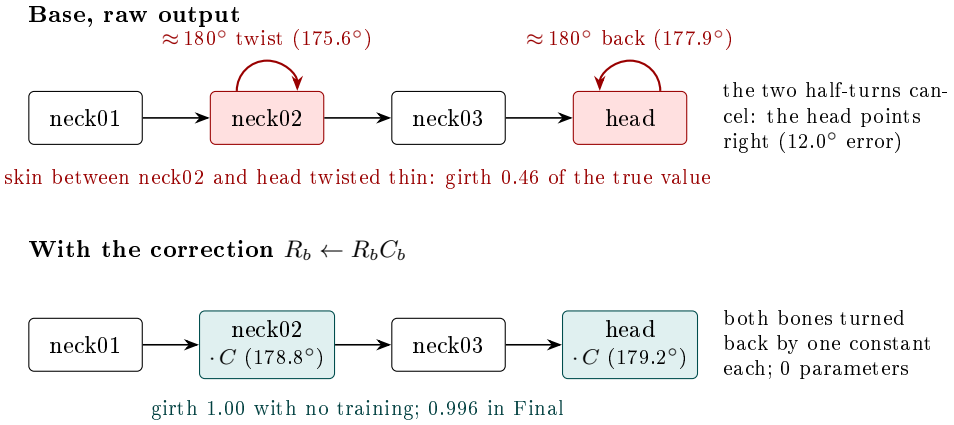


Figure 4.3.: *The neck chain neck01–neck02–neck03–head. Top: in Base, neck02 and the head bone are each turned about half a turn; the turns cancel, so the head points right, but the skin between them is twisted to 0.46 of its true girth. Bottom: one constant rotation per bone turns both back. Girth on the AnnyOne holdout, setting V1.*

How it works. The parent-relative rotation of each flipped bone b is multiplied on the right by a constant, $\mathbf{R}_b \leftarrow \mathbf{R}_b\mathbf{C}_b$. If the prediction is $\hat{\mathbf{R}}_b\mathbf{F}_b$ for the true rotation $\hat{\mathbf{R}}_b$ and a constant offset $\mathbf{F}_b$, then $\mathbf{C}_b = \mathbf{F}_b^\top$ restores the truth. I estimate $\mathbf{C}_b$ as the rotation closest to the mean of $\mathbf{R}_{b,k}^\top\hat{\mathbf{R}}_{b,k}$ over the people k of 200 random training images, never the holdout. The script selects every bone whose median error exceeds 90° with a constant offset; for Base these are neck02, head, metacarpal1.L (base of the left index finger) and finger4-1.R (right ring finger). The product comes before the helper mask, the kinematics and the losses, so training, evaluation and demos all see corrected rotations. The constants are saved with the checkpoint; the correction adds no parameters. A constant suffices because the HPH runs once from constant start values and never sees its own output.

Why not a loss, and limits. The L1 distance over matrix entries has a barrier along a twist: at 135° it is larger than at 180° (4.83 against 4.00), so a partial turn back raises the loss; a geodesic loss and the neck boost did not undo the

flip either (Section 5.4). Without training the correction worsens the global head direction, because the head bone's offset varies between people; Final therefore fine-tunes with the correction in place. A two-sided correction was not implemented, and the origin of the flip is unknown.

4.6. Training with real images

The mix. A weighted sampler draws on average 30% of each batch from 3DPW-train converted to Anny (Chapter 3; corrected labels, 5,882 images) and 70% from AnnyOne; with more labelled people per AnnyOne image, roughly 16% of the supervised people are real (an estimate). The test split is never used for training. The mix answers requirement 2a.

The hand mask. The converter fits only the root and the bones that carry a mapped SMPL joint, so every finger, thumb and metacarpal bone of a 3DPW label is at rest. With `--mask_3dpw_hands 1` the rotation terms skip these 38 bones for 3DPW people, so the fingers are learned from AnnyOne only; the mesh terms still carry the rest-pose hands.

The shape mask, tested. The converter leaves 8 phenotypes at constants (age 0.55; muscle, weight, cup size, firmness and three ethnicity values at 0.5). With `--mask_3dpw_shape 1` the shape term ignores them for 3DPW people (Final-masked); the partial variant masks only the 7 values other than age (Final-partial-mask). The mesh terms still carry the neutral body. I kept the unmasked model as Final and teacher, because both masked models lost the gender gain on real photos (Section 6.8).

4.7. Knowledge distillation

A frozen teacher supervises a smaller student on the same image, on top of the task loss (Figure 4.4).

Feature distillation. With teacher tokens $\mathbf{z}_n^t$ and the tokens $\mathbf{z}_n^s$ that the student's heads see, both of width C_t,

$$\mathcal{L}_{\text{kd}} = T^2 \cdot \frac{1}{N} \sum_{n=1}^{N} \mathrm{KL}\Big(\sigma\Big(\tfrac{\mathbf{z}_n^t}{T}\Big) \,\Big\|\, \sigma\Big(\tfrac{\mathbf{z}_n^s}{T}\Big)\Big), \tag{4.3}$$

where σ is the softmax over the channels of one token and T the temperature [HVD15]. The loss sits between backbone and heads, so the student copies the teacher's representation, as in FitNets [Rom+15].

Head transfer and output distillation. The student copies every teacher head tensor whose name and shape match. Both run on the true person locations, so person i of one is person i of the other, and

$$\mathcal{L}_{\text{kd-out}} = \alpha_{\text{rot}} \left\|\mathbf{R}^s - \mathbf{R}^t\right\|_1 + \alpha_{\text{shape}} \left\|\boldsymbol{\phi}^s - \boldsymbol{\phi}^t\right\|_1 + \alpha_{\text{dist}} \left|d^s - d^t\right| \tag{4.4}$$

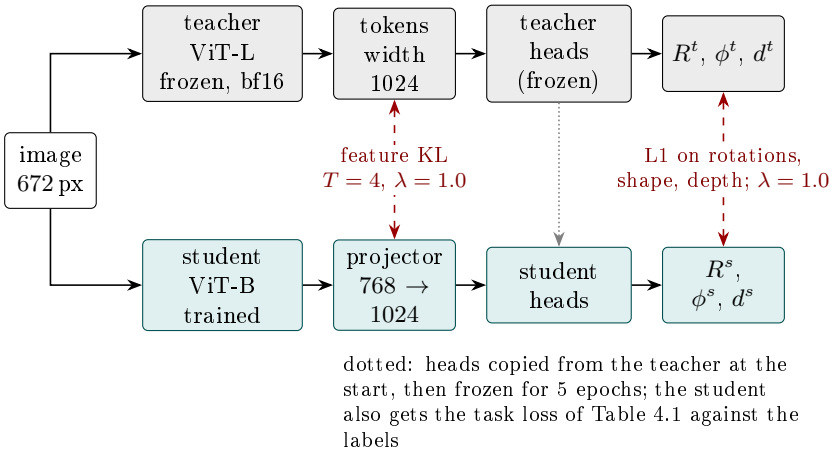


Figure 4.4.: *Distillation of the Student. The frozen ViT-L teacher (Final, backbone in bf16) and the ViT-B student see the same image. A projector maps the student's tokens from width 768 to 1024, and heads copied from the teacher read them. The feature loss compares the projected tokens with the teacher's ($T = 4$); the output loss compares rotations, phenotypes and depth of the same people. Shaded boxes train.*

pulls the student toward the teacher's detached outputs, with the task weights of Table 4.1.

Warm-up freeze and projector-fed heads. Copied heads that read through random input layers give confident wrong outputs, and their gradients damage the copied weights; the copied tensors therefore stay frozen for the first epochs. The final design removes the random input layers: a linear projector maps the student's tokens to the teacher's width, every head is built at that width (`--head_dim`), all head tensors transfer, and Equation (4.3) trains the projector toward exactly the input the heads expect. TF32 products and the teacher's backbone in bfloat16 cut a ViT-S student's step from 1.24 to 0.47 s on an A100, with unchanged evaluation numbers.

4.7.1. History: the ViT-S students

The first answer to the size requirement was a line of ViT-S students of Base, Student-S/1 to Student-S/final, trained before both fixes and the label correction (numbers in Section 6.11). They taught three lessons. Feature distillation alone leaves the student's heads unsupervised, so heads must be copied from the teacher. Copied heads need projector-fed inputs, not random input layers. And the start matters most: a backbone already trained for mesh recovery gave the largest gain, and the student was limited by the images it had seen, not by its size.

4.7.2. The ViT-B student of Final

Recipe. The Student follows the projector-fed design with a ViT-B backbone and the corrected training of this chapter (Table 4.3). Its backbone starts from NAVER's Multi-HMR ViT-B, trained for mesh recovery, and trains fully; the copied heads stay frozen for 5 epochs; the targets are exactly the teacher's. I claim no augmentation: the job script names colour jitter, but no code reads that setting. The copied heads output the half-turn that the teacher's bone correction undoes, so the student trains with that correction; afterwards its own is estimated on 200 training images and used in every evaluation.

A. Recipe		
	Final (the teacher)	Student
Backbone	ViT-L/14 (width 1024) from Base; last 4 of 24 blocks trained; frozen in bf16 when teaching	ViT-B/14 (width 768) from NAVER's HMR-trained Multi-HMR ViT-B; fully trained
Heads	HPH at width 1024 (14.01 M)	copied from the teacher, frozen 5 epochs, fed by a projector 768→1024 (0.79 M)
Params / GFLOPs / ms	318.38 M / 1,927.4 / 59.7	101.38 M / 603.3 / 28.5 (3.14× / 3.19× / 2.10× less)
Schedule	10 epochs × 1,000 iterations × batch 4 = 40,000 images	30 epochs × 1,000 iterations × batch 8 = 240,000 images
Learning rate	10^{-5}, constant	3×10^{-5}, halved every 12 epochs
Targets	head anchor, depth weight 5, 30% 3DPW (v23), hand mask	identical
Distillation	–	feature KL $\lambda = 1.0$, $T = 4$; output L1 $\lambda = 1.0$
Bone correction	four bones, estimated on Base	teacher's in training; own one afterwards
B. The 95% bars: student error ≤ reference error / 0.95		
	against Final	against Multi-HMR-Anny
AnnyOne holdout PVE (V1)	73.7 → **77.6**	60.3 → 63.5 (likely in-sample)
AnnyOne holdout MPJPE (V1)	77.9 → **82.0**	63.3 → 66.6
3DPW test PVE (v23)	89.0 → **93.7**	126.9 → 133.6
3DPW test MPJPE (v23)	88.3 → 92.9	129.4 → 136.2

Table 4.3.: *The Student (ViT-B) next to its teacher Final. Part A: recipe and measured cost. Part B: the error bars of the 95% rule in mm; the job computes the bars from the teacher re-measured in the same job, so these values are for orientation. Cost at 672 px, one person; ms on one L40S, batch 1, both models in the same job. The rule also asks for fewer parameters, fewer FLOPs and a faster forward pass. Read as "error at most 1.05 times the teacher's", the bars against Final would be 77.4 and 93.4 mm. The Student's own errors are in Chapter 6.*

The 95% rule. Requirement 1b asks for a smaller model within 95% of the large one. I read it as: student error at most the teacher's error divided by 0.95, for PVE and MPJPE on the AnnyOne holdout (V1) and the 3DPW test set, with fewer parameters, fewer FLOPs and a faster forward pass (bars in Table 4.3). NAVER's model was trained on the AnnyOne release, which includes our 100 holdout images, so its 60.3 mm is most likely in-sample; a student of a 73.7 mm teacher cannot be expected to reach its bar of 63.5 mm, while its 3DPW bar of 133.6 mm lies above the teacher's own error. Which reference counts is the supervisor's decision; Chapter 6 reports the student against both.

4.8. Evaluation protocol

Table 4.4 lists the metrics. Errors are means over matched people: a greedy 2D matcher pairs each labelled person with at most one detection, and recall is the share of labelled people that got one.

Metric	What it measures	Blind to
PVE (mm)	vertex error after both pelvises are put at the same point	placement, depth, size
PA-PVE (mm)	PVE after the best rotation, scale and shift (Procrustes)	also global orientation
MPJPE (mm)	PVE on the 163 Anny joints	as PVE
Recall, precision (%)	labelled people matched; detections matched	false detections; bystanders
Root error (m)	3D pelvis distance, no alignment	pose and shape
Size ratio	drawn head-to-pelvis distance, predicted over true (1 is right)	3D body size
Depth slope / intercept	line fitted to predicted against true pelvis depth (1 / 0 m)	errors around the line
Bone angle error (deg)	angle of $\hat{R}_b^\top R_b$ for a parent-relative rotation; $\approx$180 = flipped	compensating bones
Head direction (deg)	camera-frame head orientation error	the chain below the head
Neck girth ratio	neck-skin radius, predicted over true (1 is right)	neck length
$\mathrm{PVE}_{\mathrm{neck}}$ (mm)	vertex error over the neck skin, centred at neck01	the rest of the body
Child AUC	chance that a child gets a lower predicted age than an adult (0.5 = chance)	undetected people
Gender accuracy (%)	predicted gender on the right side of 0.5 (50 = chance)	undetected people

Table 4.4.: *Metrics used in this thesis. All errors are means over matched people; the last two are on Relative Human; MPJPE is not comparable with published 3DPW numbers. On the AnnyOne holdout the headline setting is V1 (helper bones masked); V0 leaves them unmasked and reads 3.8 mm higher in PVE on the same people. 3DPW numbers are against our converted labels only (Chapter 3).*

AnnyOne holdout. The last 100 AnnyOne images (indices 579,860 to 579,959), never trained on and without children (Chapter 3). The headline setting V1 masks the helper bones, as the demos do; V0 does not. A trace script adds the placement and per-bone metrics.

3DPW test. All 8,190 test images (11,854 people) of the corrected labels, same script for every model, setting V0. MPJPE averages all 163 Anny joints and the conversion itself errs by about 14 mm (Chapter 3), so these numbers are not comparable with published 3DPW numbers.

Relative Human and demo images. Relative Human gives real photos with an age group, a gender and a box per person. Its validation split (635 images, 2,130 people) serves for evaluation only: boxes from the projected vertices, matching at IoU ≥ 0.3, child-shaped meaning a predicted age below 0.45. The 4 demo photos and 8 3DPW test frames are rendered with each body placed from its detection box, which hides depth errors, and are judged by eye.

One run, no seed. The training code sets no random seed, and every model was trained once. Two identical pilots show that one run resolves about 10 mm PVE, 3° head direction and 2 recall points (Section 5.7); smaller gaps are within run-to-run variation. Restarting a fine-tune of Base with nothing changed already costs about 2.4 mm PVE, so every fine-tune is compared with its own control. The pelvis error has no measured noise floor, and no bootstrap intervals were computed.

5. Diagnostics: How Each Failure Was Isolated

Base showed three failures in its renders: thin, twisted necks, claw-like fingers and adult bodies on children. Before changing any training setting, I tested each belief about their causes on its own, including the ones that turned out wrong. Unless stated otherwise, every number is from Base on the 100-image AnnyOne holdout in setting V1, one run per arm without a seed, read against the variation of Section 5.7.

5.1. The three visible failures

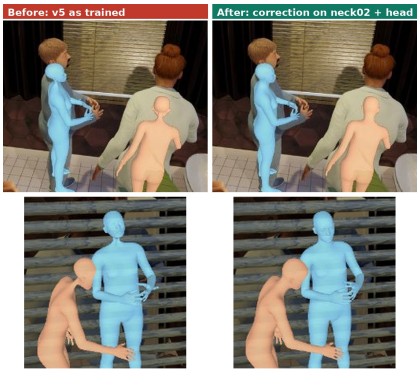


Figure 5.1.: *Holdout images 579862 (top) and 579863 (bottom). Left: Base as trained. Right: the same checkpoint with the output correction of Section 5.4, no training. Synthetic, bodies at the model's own depth, so the depth bug of Section 5.3 still draws them too small. Necks judged by eye.*

In Figure 5.1 (left) Base draws each neck as a thin stalk pinched under the head. The fingers curled into claws, so every real-photo demo set them to rest pose. A standing boy in one of our photos received an adult body (Figure 6.8).

The draft gave each failure an untested cause: skinning of the stacked neck bones ("a property of the rig"), finger resolution, and unlearned age, shown by five small people in our photos predicted at adult ages (0.57–0.67). The neck cause is wrong, resolution explains little, and the child evidence was wrong: four of the five were adults who crouched, sat or were cut off by the frame (height phenotype 0.47–0.56). Only the boy was a child, and he received age 0.60 and height 0.48: an adult. The number behind the necks of the figure is the girth of Table 5.4.

5.2. What is not the cause: six structural checks

check	sample	result	verdict
1 labels decode onto the people (no network)	manifest, 55 people	84.5 % of joints inside the image, 84.9 % inside the box (the rest is truncation); dataset camera right on 100 % of images	labels fine
2 body model, age sweep	body model	skeletal height 0.53 m at age 0, 1.55 m at age 0.8; head/height 0.200 to 0.106	child shape exists
3 single-joint moves	body model	all 16 pass (96–100 % of moved vertices in the intended part)	rig fine
4 gradients	4 holdout images	zero gradient only for the 88 helpers; all 75 active bones and 11 shape outputs get gradient (598 at the root to 4.7 at the fingers); two of three camera outputs get none	path fine
5 prediction vs. truth, shape swap	holdout, manifest	neck length 99.3 % of truth; true shape lowers the 10 manifest children's PVE 47.8 → 41.6 mm	shape head is the gap
6 capped overfit	40 manifest training images	training PVE −36 % (80.7 → 51.8 mm), holdout +45 %	learns; two warnings

Table 5.1.: *Six checks on Base before any training change. The manifest holds 40 training images (10 adults, 10 children, 10 with visible hands, 10 hard neck poses) and 15 holdout images without children; it is excluded from every later fine-tune. Check 6 trains a throw-away copy for 400 steps and reports V0.*



(a) *holdout 579888, a close-up with hands* **(b)** *training image 462901, low camera*

Figure 5.2.: *Check 1 on two manifest images: image | ground-truth mesh | joints and box, decoded through the training body model and drawn with the dataset camera; no network. By eye the mesh lies on the person, down to the fingers. In (a) the frame cuts head and legs: truncation, not a decoding error (the black corner is in the dataset image). Synthetic; two of four rendered cases.*

The checks rule out the rig, the label decoding and the gradient path. Shape dominates the loss (25.7 %; translation 1.9 %, distance 0.6 %). The cause had to be in what training asked of the network. The overfit run gave two warnings: the rotation loss barely moved while the images were memorised (44.6 → 40.8), and the translation and 2D terms rose (1.67 → 1.89, 58 → 75, first against last five epochs), possibly a first sign of the depth bug. That evidence is weak: the values are single-batch prints, and the half-run averages fall (2.10 → 1.96, 69

→ 63).

The checks are blind to two faults, and both were present: a target at another joint than the anchor (they compare pelvis-centred quantities) and a bone flipped the same way in every person (bone errors were averaged by group).

5.3. The hidden depth bug

quantity	truth	Base
pelvis depth, mean (sd)	1.92 m (0.49)	3.00 m (0.32)
people placed farther than truth	–	100 %
corr. of true depth with depth error	–	−0.76
skeletal height (3D)	1.409 m	1.405 m ($r = 0.89$)
head / pelvis 2D error, median	0	46 / 14 px
corr. of PVE with depth error	–	−0.03

Table 5.2.: *Where Base places people. V1, 100-image AnnyOne holdout, all 222 matched people; slope, intercept and drawn size are in Table 5.3. Right metric size, wrong distance; PVE does not see it.*

run (depth loss)	root (m)	size	slope	icpt. (m)	head px	pelvis px	PVE	recall
Base	1.21	0.63	0.33	2.36	64.5	23.4	70.5	90.2
2-epoch fine-tunes of Base, AnnyOne only								
control ×1	1.11	0.66	0.33	2.28	58.1	23.3	72.3	89.8
fix ×1	0.69	0.78	0.30	1.93	9.6	37.3	77.3	87.8
fix ×5	0.66	0.80	0.44	1.62	8.5	38.1	76.8	87.8
fix ×20	0.53	0.87	0.43	1.44	8.8	31.4	75.7	87.4
10-epoch fine-tunes of Base, AnnyOne only								
control ×1	1.44	0.59	0.34	2.59	64.0	24.3	72.2	89.4
fix ×1	0.73	0.78	0.30	1.98	7.6	38.5	75.5	87.0
fix ×5	0.54	0.84	0.38	1.59	7.2	33.1	74.8	87.8
10-epoch fine-tunes of Base with 30% 3DPW (v23 labels)								
Control ×1	1.40	0.59	0.28	2.66	67.1	26.7	77.4	89.0
Final ×5	0.73	0.78	0.42	1.75	7.6	41.1	73.7	87.0
other checkpoints trained with pelvis targets								
Mixed-L	1.39	0.59	0.31	2.60	–	25.1	75.0	88.6
Fresh-heads-L	1.49	0.57	0.28	2.71	–	24.0	74.1	89.0

Table 5.3.: *Placement before and after building the training targets at the head joint. V1, 100-image AnnyOne holdout, matched people; metrics as in Table 4.4, head and pelvis px = 2D error, PVE in mm, recall in %. Control rows build the targets at the pelvis, fix rows at the head; ×1/5/20: distance and translation weight. Final also has the bone correction. One run per row, no seed; each fix row is read against the control of the same length (none was run at ×5 or ×20).*

Base places every person too far (Table 5.2, Figure 6.2), the nearest farthest. The skeleton has the right height, so the body is drawn at 0.63 of its true size.

PVE is blind to all of it.

The code gave the cause: the mesh is placed by its head joint, but the anchor targets were built at the pelvis (Section 4.4). A one-batch debug print showed offsets of -2 to -19 cells where the legal range is ± 0.5. The likely mechanism, not tested on its own: with the mesh hanging from a misplaced head, the 2D terms can shrink the misfit only by drawing the body smaller, that is, farther away, and the direct depth terms (2.5 % of the loss) could not resist; near people have the largest misfit.

Table 5.3 tests the fix against same-schedule controls. All five fix runs reach a root error of 0.53–0.73 m, both controls stay at 1.11–1.44 m, the drawn size rises from 0.59–0.66 to 0.78–0.87, and the head lands within 7–10 instead of 58–67 pixels: every fix run lies below every control, far outside run-to-run variation. Two other checkpoints trained with pelvis targets carry the bug too. It stayed hidden because PVE aligns the pelvis first and the real-photo demos place each body from its detection box.

Not shown is correct depth: the slope barely moved (0.30 with the switch alone, 0.38–0.44 with a stronger depth loss), and predictions are still pulled toward 2.5–2.9 m for a reason I have not found. The pelvis 2D error rose to 31–41 pixels, likely because the head is now pinned while the body is still too small (not checked). Every fix run is worse than its control by $+2.6$ to $+4.9$ mm PVE and -1.6 to -2.4 recall points: each gap within variation, all in one direction, so a likely cost. By the success rule written beforehand (PVE ≤ 71.5 mm, recall ≥ 89.2 %) every run failed, the controls included.

5.4. The neck: a per-bone dump

A per-bone dump of the neck chain found the fault (Table 5.4, first column). neck02 is 175.6 deg wrong, a half-turn about its own axis (twist 175.7, swing 3.3 deg), and the head bone 177.9 deg; neck03 is right relative to its parent (5.7 deg) but 173.3 deg off in the camera frame. The half-turns cancel, so the head points the right way (12.0 deg); the skin between is twisted thin. Group means (62.7 and 89.5 deg) had hidden the flip. It is in all eight checkpoints of the Base line (neck02 175.0–176.0 deg, girth 0.46–0.47), from the first stage on, before the helper mask, the neck boost and the shape weight of 20; the depth pilot keeps it, so depth is not the neck's cause. Fresh-heads-L has a normal neck (girth 1.00) but flips other bones: a compensated pair in the left arm and three metacarpals.

This refutes the draft: the rig passes all 16 single-joint moves, and rest pose (all rotations identity) hides the flip. A loss did not undo it: the neck boost ($\times 2.41$ for the boosted bones after renormalisation) left it unchanged, and a geodesic loss moved neck02 only from 175.6 to 175.0 deg, likely because neck02 and head must turn back together. So I corrected the output with one constant rotation per flipped bone (Section 4.5): 178.8, 179.2, 179.8 and 179.7 deg for neck02, head, metacarpal1.L and finger4-1.R.

Without training the correction restores neck02, girth and neck-skin error, but the head direction worsens to 19.1 deg, above the 15-degree limit set be-

	Base	corr., no train.	corr. + 2-ep. FT	2-ep. control	Control 10 ep.	Final 10 ep.
neck02 (deg)	175.6	4.5	4.7	175.7	175.8	4.8
head bone (deg)	177.9	12.4	11.2	177.9	177.7	8.9
metacarpal1.L (deg)	180.0	0.0	0.0	–	180.0	0.0
finger4-1.R (deg)	n.d.	10.2	10.3	–	180.0	9.6
neck girth, pred./true	0.461	1.004	0.997	0.461	0.464	0.996
PVE_{neck} (mm)	27.3	10.8	10.7	27.9	27.7	11.0
body PVE (mm)	70.5	71.3	68.8	76.9	77.4	73.7
PA-PVE (mm)	53.3	52.2	50.9	55.2	56.4	50.3
recall (%)	90.2	90.2	89.4	91.1	89.0	87.0
head direction (deg)	12.0	19.1	14.4	13.8	13.0	13.1
fingers (deg)	17.3	8.2	8.3	17.4	17.4	8.3

Table 5.4.: *The half-turned bones and their correction. V1, 100-image AnnyOne holdout, matched people. Bone errors: local medians (about 180 deg = half a turn). PVE_{neck} over the 171 neck-skin vertices. Corr.: the bone correction (four constants from 200 training images); FT: fine-tune. 2-epoch columns: AnnyOne-only fine-tunes of Base; Control and Final: 10 epochs with 30% 3DPW. n.d.: not dumped. One run per column, no seed; PVE gaps below about 10 mm are within run-to-run variation (Table 5.7).*

forehand, likely because the head's error differs per person; a 2-epoch fine-tune brings it to 14.4 deg (control 13.8). The body-PVE gaps lie within variation. Figure D.2 shows two more holdout pairs. Not shown: why the bones flipped (the singular 6D start value fits the tilted metacarpal1.L, not neck02's twist), and necks on real photos, which have no ground truth (Figure 6.3, by eye).

5.5. The fingers

In Table 5.5 predicted fingers are 17.3 deg off and fingers at rest 15.1 deg: the prediction was worse than none. Hands are small (2.25 tokens long at the median, under two tokens for 41 %), but a person crop at test time leaves the finger error at 17.2 deg while body PVE nearly doubles, and a region-normalised rotation loss (a rebalance from about 70 to 33 % of the rotation loss, not an upweighting) changed nothing. The per-bone search on training images then found metacarpal1.L and finger4-1.R half-turned; correcting them roughly halves the finger error, with and without training.

About half the finger error was two flipped bones, not resolution. A hand-crop branch was never tried, and a rest-pose floor with the correction was not measured. Fingers on real photos: Figure 6.4.

5.6. Children and body shape

Table 5.6(a) is the draft's shape-weight test on 10 holdout images: raising the shape weight from 1 to 20 moved the age bias from +0.199 to −0.002 and the

(a) finger rotation error on the holdout

setting	fingers (deg)	reference (deg)
Base, predicted fingers	17.3	–
Base, fingers set to rest pose	15.1	–
Base, test-time person crop (PVE 124.8 mm)	17.2	17.3 (paired)
region-normalised rotation loss, 2 ep.	17.5	17.4 (control)
Base + bone correction, no training	8.2	17.3 (Base)
Base + bone correction, 2-ep. fine-tune	8.3	17.4 (control)
Final (10 ep., 30% 3DPW)	8.3	17.4 (Control)
Fresh-heads-L, three metacarpals corrected	17.8	32.0 (uncorr.)

(b) hand size in tokens (14-px patches of the 672-px input)

	median length	under 2 tokens
full image, all hands	2.25	41%
person crop at test time	4.2	0%
children's hands, full image	0.7	80%

Table 5.5.: *Fingers. (a) Finger rotation error, V1, 100-image AnnyOne holdout, matched people, against the matching run without the change; rest pose keeps the flipped metacarpal1.L. (b) Hand length in input tokens. One run per row, no seed.*

(a) predicted minus true phenotype, 10 holdout images (draft measurement)

phenotype	true mean	true std	weight 1	weight 20 (Base)	reading
age	0.553	0.036	+0.199	−0.002	near the mode
height	0.579	0.026	−0.154	−0.026	near the mode
proportions	0.763	0.070	−0.017	−0.001	near the mode
firmness	0.757	0.039	−0.167	−0.190	unchanged
gender	0.694	–	−0.042	−0.063	≈
weight	0.382	–	+0.056	+0.064	≈

(b) Base on the full holdout and in the young tail

holdout, all 222 people (nobody younger than 0.513)	age bias −0.015 (abs. 0.037); gender −0.008 (0.068)
holdout, true age 0.6–0.8 (n = 53, mean 0.67)	predicted age 0.57
50 youngest training people (in-sample, true age 0.015)	predicted age 0.10; height −0.25
training medians (AnnyOne)	gender 0.588, age 0.601, height 0.500

Table 5.6.: *Shape head, phenotype units* $[0, 1]$*. (a) The stage before Base (shape weight 1) against Base (weight 20) on the same 10 holdout images. (b) Base, V1, on the full holdout and the 50 youngest training people (in-sample). In both tails predictions move toward the training median.*

height bias from −0.154 to −0.026. Part (b) shows that this holds near the mode only: in both tails predictions move toward the training median (the holdout has no children, so the young tail is in-sample). On Relative Human (Table 6.2) Base reads children as short adults: kids get a mean age of 0.61, babies 0.62 and adults 0.63 (AUC 0.61), none of 89 kids and 1 of 38 babies fall below the child threshold; the smallness goes into height instead. Gender

carries no signal (accuracy 46.5 %). On our demo photos the medians of all 299 shape lines (gender 0.59, age 0.61, height 0.50) equal the training medians.

Children and female-looking men are one failure: with weak image cues the shape head returns the training median. Two mechanisms fit, and the evidence cannot rank them: the gap between synthetic and real images, or the loss (an L1 on sigmoid outputs regresses to the median, the sigmoid's gradient is 0.25 at 0.5 but 0.015 at 0.015, and only 2.53 % of training people are below age 0.3). It looks like a short adult because Anny's child shape (shoulder over head width 2.24 at age 0.1, 3.00 at 0.6) appears only below an age of about 0.4–0.5: a low height at age 0.6 gives a short adult (at least 1.22 m), and proportions barely matter (leg over stature −0.016).

Not shown: which mechanism dominates, and ages in years (the code uses three age scales that disagree).

5.7. Pilot noise and restart cost

quantity	measured	one run resolves
two identical controls: PVE	76.3 vs. 68.7 mm (7.6 mm)	about 10 mm
head direction	13.6 vs. 11.7 deg (1.9 deg)	about 3 deg
recall	0.8 points apart	about 2 points
six arms, Base recipe	PVE 68.7–76.3 mm, recall 88.6–90.2 %, head 11.3–13.7 deg	
restart cost	72.9 vs. 70.5 mm for Base (2.4 mm)	read against own control
root error	no measured floor	fix 0.53–0.73, control 1.11–1.44 m

Table 5.7.: *Run-to-run variation of the 2-epoch pilots (fine-tunes of Base at learning rate 10^{-5}, no seed). V1, 100-image AnnyOne holdout, matched people. Restart cost: what two more epochs of fine-tuning Base cost with nothing changed.*

By Table 5.7 one pilot resolves about 10 mm PVE, 3 deg and 2 recall points, and is read against its own control because of the restart cost. That suffices for every change this chapter relies on (root error, bone angles 175 to 5 deg, girth 0.46 to 1.00, fingers 17.3 to 8.3 deg), but not for a PVE change of a few millimetres, so the PVE costs of the depth fix and the neck pilot stay unresolved; no bootstrap intervals were computed.

5.8. What did not work

Six changes did not fix a failure (Appendix E, Table E.1): an age-balanced sampler (inconclusive on 10 in-sample children), a global head orientation loss (within the spread of two identical controls), a region-normalised hand loss, a geodesic rotation loss, the neck and hand boost, and a ×20 depth loss (no gain over ×5). Each worked around a fault the loss could not see; the two fixes that worked came from the diagnostics.

6. Experiments and Results

This chapter reports what the models built on the diagnostics of Chapter 5 achieve, in the order the fixes were made. Each section opens with its figure or table, reads it, interprets it and says what it does not show.

6.1. Evaluation sets and settings

Four sets are used, all described in Chapter 3 and scored as in Section 4.8. The AnnyOne holdout holds the last 100 AnnyOne images (222 people, no children), never trained on; its headline setting V1 masks the helper bones, and a trace script adds the placement metrics (root error, depth fit, drawn size) with each body at the model's own predicted depth. The 3DPW test set is all 8,190 test images of the corrected labels, scored in setting V0 with one script for every model; its numbers hold only against our converted labels, which err by about 14 mm, and are not comparable with published 3DPW numbers. The Relative Human validation split gives real photos with an age group, a gender and a box per person, but no 3D body. The 12 demo images (4 photos and 8 3DPW test frames) have no labels; they are judged by eye, with bodies placed from the detection box.

The models are those of Table 4.2. The central comparison is Final against Control: the two runs share the start point (Base), the schedule, the real-image share and the hand mask, and differ only by the depth fix and the bone correction. Every model was trained once, without a fixed seed. Two identical runs differed by 7.6 mm PVE, 1.9 deg head direction and about 2 recall points (Section 5.7); I call gaps below these "within run-to-run variation" and compare every fine-tune with its own control, because restarting a fine-tune of Base already costs about 2.4 mm PVE. Setting V0 reads about 3.8 mm higher than V1.

6.2. Depth: the anchor fix on the holdout

Figure 6.1 shows the bug and the fix on one image: Base draws the man far behind his true position and too small, the depth pilot close to his true size. Figure 6.2 shows all people. Control places 99.1 % of them too far. Final pulls the cloud down: the root error falls by 48 % (1,402 to 728 mm), the drawn size rises from 0.59 to 0.78, the depth correlation from 0.46 to 0.56, and the head lands within 7.6 pixels of its true position against 67.1 (Table 6.1a).

The fix survives ten epochs with real images and the bone correction: every run with head targets, pilots included (Table 5.3), lies below every run without them. What moved is the offset and the size, not the slope. Final's fit crosses

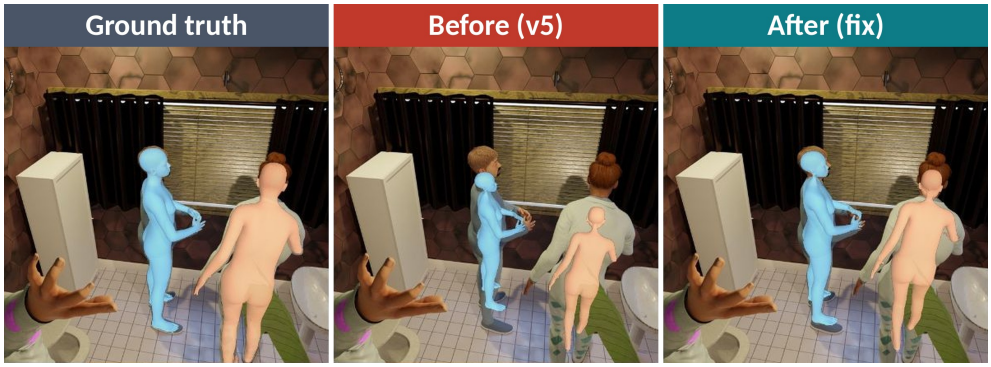


Figure 6.1.: *Holdout image 579862 rendered at each model's own predicted depth: ground truth, Base, and the depth pilot (head-joint targets, depth weight ×5, 10 epochs, AnnyOne only). The predicted body goes from 0.76 to 0.95 of its true height. Synthetic image; a good case, not a typical one. Image 579860 is in Figure D.1.*

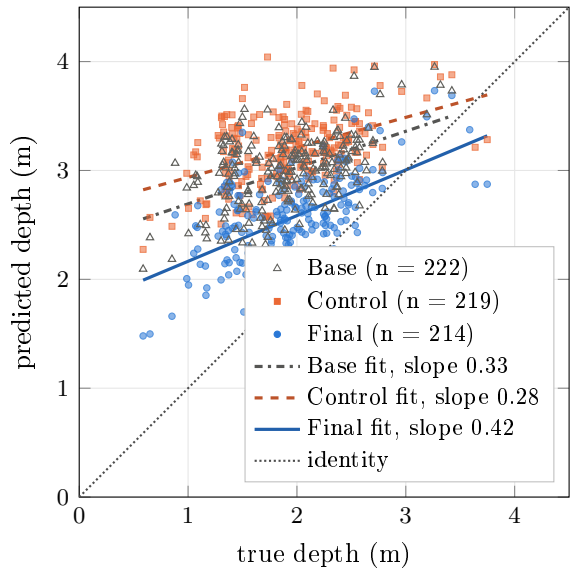


Figure 6.2.: *Predicted against true root depth for every matched person of the AnnyOne holdout (100 images, setting V1, each body at its own predicted depth): Base (222 people), Control (219) and Final (214). Lines: least-squares fits and the identity. Final's fit is* $0.42 \times \text{true} + 1.75$ *m, Control's* $0.28 \times \text{true} + 2.66$ *m, Base's* $0.33 \times \text{true} + 2.36$ *m. One run per model; synthetic images. Statistics in* `data/plot_stats.md`.

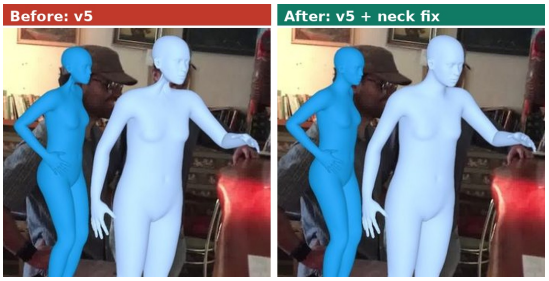


Figure 6.3.: *Real 3DPW test frame* `downtown_bar_00` *701: Base (left) and the neck pilot, fine-tuned 2 epochs with the correction (right); necks ringed. Bodies placed from the detector box; judged by eye. The columns differ by the correction and the fine-tune. A second frame is in Figure D.3.*

the identity at 3.01 m, so near people are still placed too far (94.4 % of all people). The pelvis lands farther from its pixel (41.1 against 26.7 pixels), likely because the head is pinned while the body is still too small; I did not check this.

Not shown: correct near and far depth, and why the slope does not move. Final's root error is above the 10-epoch depth pilot's (544 mm); the real images or the bone correction may cause this, which was not tested. All depth numbers are synthetic, and Figure 6.1 shows the pilot, not Final.

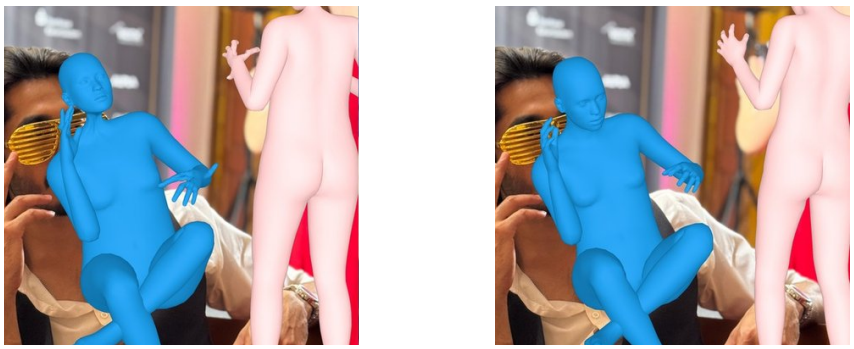
6.3. The bone correction

Figure 6.3 shows the correction on a real frame that no model trained on: Base draws both necks as thin stalks under the head, the corrected model draws them at normal width. Figure 5.1 and Table 5.4 give the correction without training and after a 2-epoch fine-tune. Final keeps it through ten epochs with real images (Table 6.1a): neck02 is 4.8 deg off against 175.8 for Control, the neck girth 0.996 of the truth against 0.464, and the joint error of the neck region falls from 62.6 to 5.5 deg. The head direction, which got worse with the correction alone (19.1 deg), is back at Control's level.

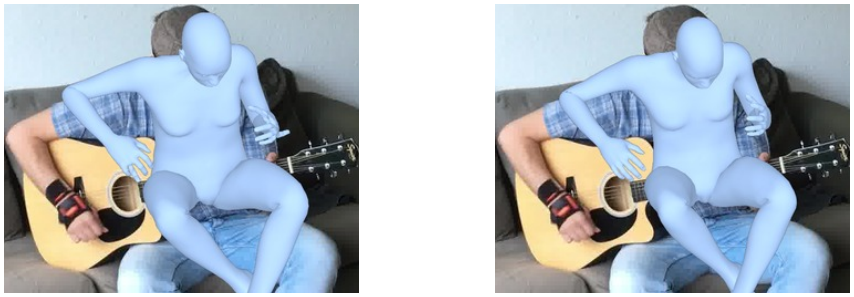
The half-turn does not come back during training: the network learns on top of the correction instead of undoing it. The correction is one constant rotation per bone, adds no parameters and says nothing about why the bones flipped. PVE shows no sign of a cost (Section 6.5).

Not shown: a neck number on real photos, which have no neck ground truth. The real frame shows the neck pilot, not Final, and its columns also differ by the fine-tune; on the holdout the same fine-tune without the correction keeps girth 0.461 (Table 5.4), so the correction is the likely cause of the wider necks. Flips that compensate each other in a chain (Section 5.4) need a two-sided correction, which I did not implement.

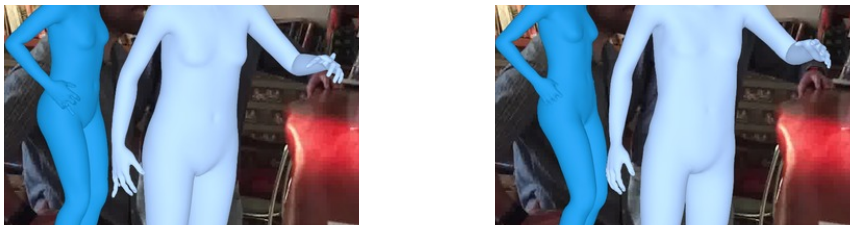
Base, predicted fingers | Final, predicted fingers



(a) party photo: the hand at the face and the hand over the armrest



(b) 3DPW test frame, guitar: strumming hand (left) and fretting hand (right)



(c) 3DPW test frame `downtown_bar`: the hand on the bar and the hanging hand

Figure 6.4.: *Predicted fingers (not set to rest pose) on three demo images, zoomed on the hands; Base left, Final right. Bodies placed from the detector box, so meshes sit beside their people. Base still carries the two flipped hand bones; Final has the correction. Neither model puts the fingers on the real ones. Three images, judged by eye; the columns differ by the correction and the fine-tune. All twelve montages are in Figure D.7.*

6.4. Fingers

Predicted fingers are 17.3 deg off in Base and 17.4 deg in Control, worse than fingers left at rest (15.1 deg; Table 5.5). Final, with the two flipped hand bones corrected, reaches 8.3 deg (Table 6.1a); the correction alone, without training, gave 8.2.

About half of the finger error was two flipped bones, metacarpal1.L and finger4-1.R, and ten epochs of training keep the gain. The rest is not explained. Real images add nothing to the fingers: the hand mask removes the rotation

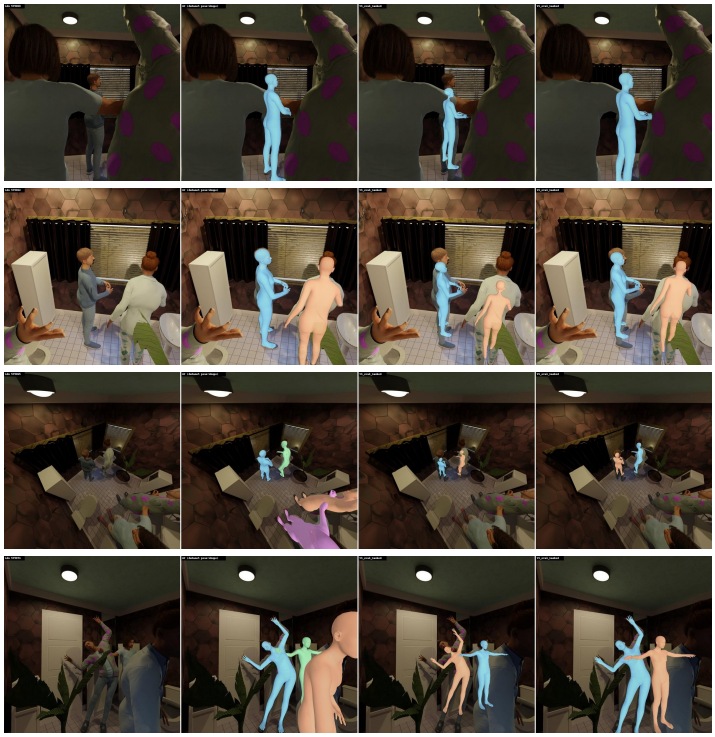


Figure 6.5.: *Four AnnyOne holdout images (index in the first panel), each as image | ground truth | Base | Final. Ground truth drawn with the dataset camera; Base and Final in setting V1 (panels labelled "V1_eval_masked"), each body at its model's own predicted depth. By eye, Base draws its bodies smaller and further back than the ground truth, Final close to its size and place. Mesh colours mark detections, not identities. People without a ground-truth mesh (the foreground arm in 579862) are not counted; people a model misses (the two lying in the foreground of 579865) have no mesh. Synthetic images, one run per model.*

terms of the 38 hand bones for 3DPW people, whose labels hold rest-pose hands.

Figure 6.4 shows predicted fingers on real images; every earlier demo set them to rest pose. By eye, Base's fingers are splayed and stiff, with one finger of the left hand sticking out as the flipped metacarpal1.L would make it, while Final's are closed and bent and read as hands: draped over an armrest, curled as if fretting, bent over a bar edge. Where the hands are small (Figure D.7), nothing can be judged at print size.

Not shown: a finger number on real images. The 3DPW labels carry rest-pose hands and the demo images have no labels, so the real-image evidence is three images by eye, showing the correction and the fine-tune together.

	Base	Depth pilot	Neck pilot	Control	Final
PVE (mm)	70.5	74.8	68.8	77.4	73.7
PA-PVE (mm)	53.3	54.0	50.9	56.4	**50.3**
Recall (%)	90.2	87.8	89.4	89.0	87.0
Root error (mm)	1208	544	1429	1402	**728**
Depth slope	0.33	0.38	0.34	0.28	0.42
Depth intercept (m)	2.36	1.59	2.56	2.66	1.75
Size ratio	0.63	0.84	0.58	0.59	0.78
Neck girth pred/GT	0.461	0.457	0.997	0.464	0.996
$\mathrm{PVE}_{\mathrm{neck}}$ (mm)	27.3	27.7	10.7	27.7	11.0
Global head direction (deg)	12.0	12.7	14.4	13.0	13.1
Fingers (deg)	17.3	17.4	8.3	17.4	8.3
neck02 bone (deg)	175.6	175.5	4.7	175.8	4.8
head bone (deg)	177.9	178.0	11.2	177.7	8.9

(a) *AnnyOne holdout, 100 images, setting V1 (helper bones masked, matched people); root error, depth fit and size ratio with each body at the model's own predicted depth.*

	Base	Mixed-L	Neck pilot	Control	Final	Multi-HMR-Anny
PVE (mm)	121.8	95.5	120.2	91.5	**89.0**	126.9
PA-PVE (mm)	72.0	62.9	68.7	59.7	**57.6**	68.8
MPJPE (mm)[b]	126.9	94.8	124.8	90.9	**88.3**	129.4
Precision (%)	54	73	53	62	64	75
Recall (%)	100	94	100	99	99	97
AnnyOne test-100 PVE[c]	74.3	78.5	72.7	81.2	77.5	(60.3)[a]

(b) *3DPW test, corrected (v23) labels, all 8,190 images, setting V0, same evaluation script for every model. Mixed-L was trained on the labels before their correction and is tested on the corrected ones, like every model here.*

Table 6.1.: *Final against its Control and against Base. One run per model, no seed; two identical runs differed by 7.6mm PVE, so the PVE gap between Control and Final (77.4 vs 73.7) is within run-to-run variation, while PA-PVE, root error, neck and fingers are not. Depth pilot: depth fix, depth weight ×5, 10 epochs; neck pilot: bone correction and a 2-epoch fine-tune. 3DPW numbers are against our converted labels and are not comparable with published 3DPW results.* [a]*Multi-HMR-Anny masks the helper bones by design, so its number is V1 and compares with Final's 73.7 in (a); it was most likely trained on these 100 images.* [b]*Mean over all 163 Anny joints, pelvis-centred.* [c]*The same 100 images as (a) in setting V0 (helper bones unmasked).*



(a) `downtown_bar` (b) *weekly market*

(c) `office_phoneCall` (d) *guitar*

Figure 6.6.: *Four 3DPW test frames, each shown as photo and Base (left half), then photo and Final (right half; the labels in the image name the two checkpoints). Bodies placed from the detector box, fingers at rest pose; judged by eye, no metric. The other four demo frames are in Figure D.4 (Appendix D).*

6.5. Final against Control and Base on AnnyOne

Table 6.1a puts Final next to Control, Base and the two pilots. Final has the lowest PA-PVE of every model, 6.1 mm below Control. Its PVE is 3.7 mm below Control's, within run-to-run variation, and Base keeps the lowest PVE, as it did against every fine-tune. Recall is 2.0 points below Control, at the limit of the variation, and 3.2 below Base. Figure 6.5 shows Base's bodies small and off their people and Final's close to the ground truth. The in-training curves (Figure D.6, Appendix D) put Final below Control at all ten epochs on 3DPW validation, while on the in-training holdout (V0) the order changes from epoch to epoch.

The fixes cost nothing in body error that one run can resolve, and they bring the largest PA-PVE gain of the project. PVE aligns the pelvis, so it cannot see the depth gain, and the neck and fingers are few vertices of a whole body (Section 7.2). The recall cost against Base is larger than the variation and is real. A likely reason is that the detection cell moved from the pelvis to the head, so people with a hidden head are found less often; this was not tested.

Not shown: PVE paired on the same people (the matched sets differ: 214, 219 and 222 people), recall on people with a visible head, bootstrap intervals, and a second run of Final.

6.6. 3DPW test on the corrected labels

Table 6.1b scores every model on all 8,190 test images with the same script. Final has the lowest error on our 3DPW-Anny labels, 89.0 mm PVE, ahead of

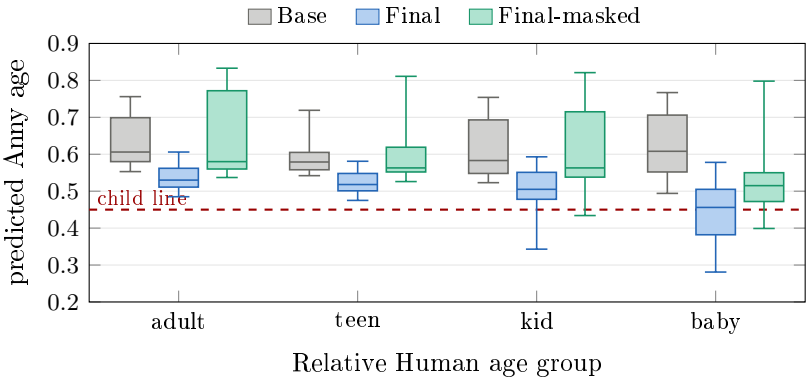


Figure 6.7.: *Predicted Anny age per Relative Human age group for Base, Final and Final-masked, matched (detected) people of the validation split only. Box: quartiles and median; whiskers: 5th and 95th percentile. Dashed: the child line, age 0.45. Ages are Anny phenotype values, not years. One run per model; statistics in* `data/plot_stats.md`.

Control, Mixed-L (the previous best), Base and Multi-HMR-Anny, and finds 99 % of the labelled people. Precision is low for every model. By eye (Figure 6.6), most men get male bodies in Final and female-looking ones in Base, Final's bodies are a little larger, and the seated guitar player is poor in both.

Most of the gain over Base comes from training on 3DPW at all: Control, without either fix, is 30.3 mm better than Base. The fixes add 2.5 mm PVE and 2.1 mm PA-PVE on top, within run-to-run variation for PVE. Final is 6.5 mm better than Mixed-L, which saw the old labels; whether the corrected labels cause part of this was not tested, since Mixed-L also differs in schedule. Precision is low because 3DPW labels only one or two actors per sequence, so every detected bystander counts as a false positive, as do ghost bodies on half-visible people.

Not shown: a comparison with published 3DPW results, which these numbers against converted Anny labels do not allow; a breakdown by sequence; and more than 8 frames by eye.

6.7. Real photos: children and gender

Table 6.2 measures body shape on real photos with labels. Final finds more people than Base and gets the gender right for 64.2 % of them against 46.5 %. Women are recognised (mean predicted gender 0.861, where 1 is female); men stay in the middle (0.551). Children separate from adults: the AUC rises from 0.606 to 0.719, and 49.3 % of the babies are child-shaped against 2.6 % for Base. In Figure 6.7 Base's boxes stand at the same height for every group, Final's step down from adults to babies, and only the baby box reaches below the child line. On the demo photos (Figures 6.8 and D.5) most men look male in Final and the boy keeps an adult body.

The AUC ranks people by predicted age, so a common shift of all ages

	Base	Final	Final-masked	Final-partial-mask
Recall (%)	47.1	64.7	57.2	63.0
Gender accuracy (%)	46.5	64.2	49.3	47.4
mean gender, men (male = 0)	0.535	0.551	0.717	0.725
mean gender, women (female = 1)	0.551	0.861	0.860	0.888
Child-vs-adult AUC	0.606	0.719	0.708	0.727
Mean age, adult	0.631	0.536	0.638	0.532
teen	0.597	0.524	0.616	0.515
kid	0.613	0.500	0.611	0.506
baby	0.624	0.438	0.524	0.465
Babies with age < 0.45 (%)	2.6	49.3	16.9	34.3
Mean height, adult	0.507	0.500	0.505	0.527
kid	0.451	0.466	0.461	0.497
baby	0.394	0.464	0.478	0.504
Matched children + adults	948	1,305	1,157	1,270
Demo images: standard deviation of the prediction				
Muscle	–	0.04	0.06	n/m
Weight	–	0.04	0.11	n/m
Cup size	–	0.05	0.11	n/m
Firmness	–	0.04	0.11	n/m
Age	–	0.03	0.08	n/m

Table 6.2.: *Body shape on real photos. Upper part: Relative Human validation split (635 images, 2,130 labelled people), detected people only, matched to the labelled boxes at IoU $\geq$ 0.3; ages and heights are Anny phenotype values in $[0, 1]$, not years; "child-shaped" means a predicted age below 0.45; the AUC ranks kids and babies against adults by predicted age (0.5 = chance). The matched sets differ between models because recall differs. Lower part: spread of the predicted phenotypes over all detected people of the 12 demo images (4 photos and 8 3DPW test frames), not matched to people; not reported for Base, not measured (n/m) for Final-partial-mask. Final-masked masks all eight constant 3DPW phenotypes, Final-partial-mask the seven at 0.5 (age kept supervised; Section 6.8). One run per model, no seed. The neck pilot (bone correction and a 2-epoch fine-tune) is the same as Base within run-to-run variation (AUC 0.613).*

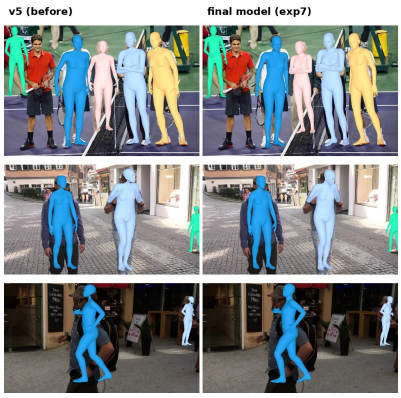


Figure 6.8.: *Three real photos never used in training (a tennis photo and the 3DPW test frames `downtown_walking_00` and `downtown_runForBus_00`): Base left, Final right, bodies placed from the detector box. Most men get male bodies, women stay female, the boy stays an adult body. Judged by eye; no metric exists for these images.*

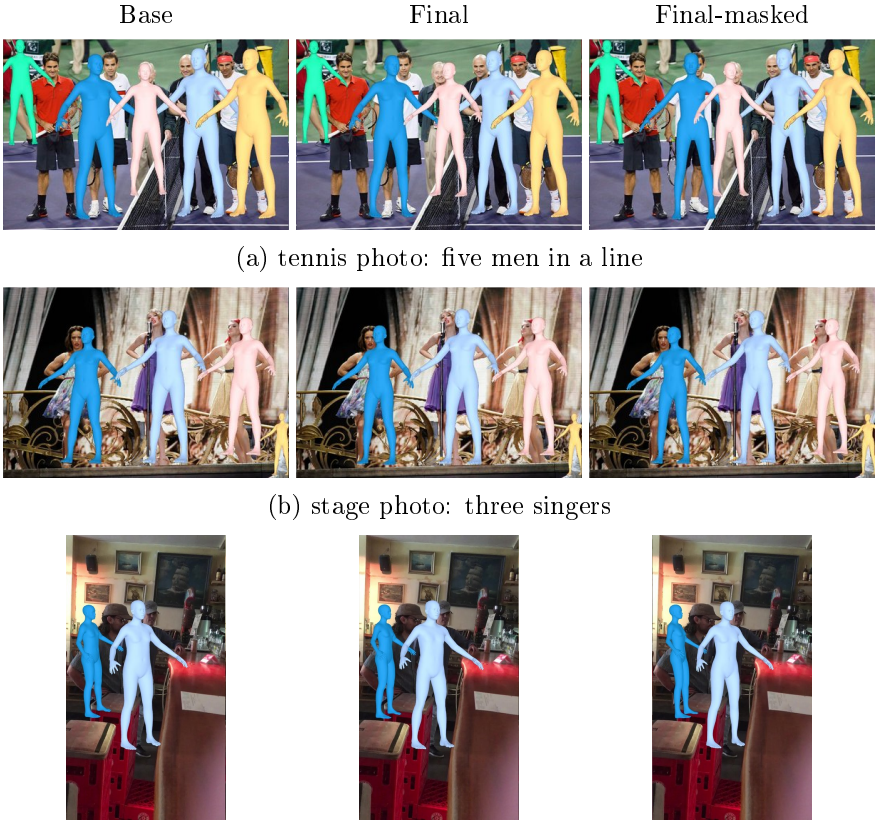
cannot raise it: the step is real separation of children from adults. Part of the lower ages is likely such a shift from the 3DPW age label, which the converter fixes at 0.55 for every real person. The gender gain fits the corrected 3DPW gender labels, which supervise the shape head on real people for the first time. Children are a measured improvement, not solved, and men are not yet male.

Not shown: ages in years (the code uses three age scales that disagree), children on held-out synthetic images (the holdout has none), and any shape number beyond age, gender and height. The numbers cover detected people only, so the matched sets differ between models.

The rest-pose line-up of Figure 6.9 draws every detected person in rest pose, so only the body differs. By eye, Base gives nearly everyone the same slim, flat body. Final gives the singers a fuller chest and hips and the man in the bar a broader torso. The tennis men keep a slim body with a slight chest in all three models, in line with the men's mid-range gender score. Final-masked looks like Final except for a slimmer man in the bar.

6.8. The shape-mask trade-off

Two runs test the shape mask of Section 4.6, one run each. Final-masked stops the shape loss on the 8 values the converter fixes for every 3DPW person: age 0.55 and seven values at 0.5. Final-partial-mask stops it only on the seven and keeps age supervised. Gender stays supervised in both. Table 6.3 sets the two next to Final. On Relative Human, Final-masked loses most of Final's gain (gender 49.3 %, babies child-shaped 16.9 %), but on the demo images it brings back variety: the spread of weight, cup size and firmness rises from 0.04–0.05 to 0.11. Final-partial-mask keeps the children (AUC 0.727, 34.3 % of babies child-shaped) and loses gender as the full mask does, with both sexes on the female side (men 0.725, women 0.888).



(a) tennis photo: five men in a line

(b) stage photo: three singers

(c) 3DPW test frame `downtown_bar`: two men

Figure 6.9.: *Rest-pose line-up on three demo images: every detected person drawn in rest pose over the photo, so only body shape differs; Base, Final and Final-masked from left to right. Bodies placed from the detector box; judged by eye, one run per model. The other nine images are in Figure D.8.*

	Final	Final-masked	Final-partial-mask	Rule
Masked 3DPW phenotypes	none	8 (age + 7)	7 (age kept)	
AnnyOne holdout, V1				
PVE (mm)	73.7	71.7	73.0	
PA-PVE (mm)	50.3	50.7	51.2	
Recall (%)	87.0	86.2	87.4	
Root error (mm)	728	682	680	
3DPW test, v23 labels, V0				
PVE (mm)	89.0	91.1	89.9	
PA-PVE (mm)	57.6	59.0	58.9	
Relative Human validation				
Recall (%)	64.7	57.2	63.0	
Gender accuracy (%)	64.2	49.3	47.4	≥ 60
Child-vs-adult AUC	0.719	0.708	0.727	≥ 0.7
Babies with age < 0.45 (%)	49.3	16.9	34.3	
Demo images, standard deviation of the prediction				
Muscle	0.04	0.06	n/m	> 0.08
Weight	0.04	0.11	n/m	> 0.08

Table 6.3.: *The three shape-mask arms. Final supervises all converted 3DPW phenotypes; Final-masked masks the 8 constants of the converted labels (age 0.55 and seven values at 0.5); Final-partial-mask masks only the seven at 0.5 and keeps age supervised. Gender stays supervised in all three. AnnyOne holdout: 100 images, setting V1; 3DPW test: all 8,190 images, corrected labels, setting V0; Relative Human: validation split, matched people (full columns in Table 6.2); demo: 12 images, not matched to people; n/m: not measured. Rule: the thresholds, fixed before the first masked run, that a masked arm had to meet to replace Final. One run per arm, no seed; two identical runs differed by 7.6 mm PVE.*

	Cost				AnnyOne holdout (V1)			3DPW test (v23)			
Model	Params (M)	GFLOPs	ms A100	ms L40S	PVE	PA-PVE	Recall	PVE	PA-PVE	Recall	Trained on 3DPW
Multi-HMR-Anny (NAVER)	380.94	$\geq$1917.2[b]	n/m	n/m	60.3[a]	44.6[a]	89.0[a]	126.9	68.8	97	no
Final	318.38	1927.4	77.9	59.7	73.7	50.3	87.0	**89.0**	**57.6**	99	yes
Control	same network as Final[c]				77.4	56.4	89.0	91.5	59.7	99	yes
Base	same network as Final[c]				70.5	53.3	90.2	121.8	72.0	100	no
Student (ViT-B)	101.38	603.3	n/m	28.5	119.6	80.8	85.4	110.1	71.8	99	yes
Backbone only, analytic count											
DINOv2 ViT-S/14	22.06	196.9									
DINOv2 ViT-B/14	86.58	589.5									
DINOv2 ViT-L/14	304.37	1917.2									

Table 6.4.: *Accuracy and cost; one run per model, no seed. 3DPW numbers are against our converted labels and not comparable with published 3DPW results. GFLOPs count one multiply-add as two FLOPs, at 672 px input with one person; latency is the median forward pass at batch 1 (fp32 with TF32). The two GPUs differ, so only times in the same column compare: the Student is 2.10× faster than Final on the L40S and has 3.14× fewer parameters and 3.19× fewer FLOPs. n/m: not measured.* [a]*Most likely in-sample for Multi-HMR-Anny, which was trained on the AnnyOne release; ours are held out.* [b]*Backbone only: the head's FLOPs and the latency cannot be counted for NAVER's model class; its head has 76.57 M parameters against 14.01 M in ours.* [c]*Base, Control and Final share the architecture; the bone correction adds no parameters.*

The body errors of both arms stay within run-to-run variation of Final, since the 3DPW meshes keep the neutral body; the trade-off is in the shape head. Without the mask, Final learned the seven constants and predicts nearly the same muscle, weight, cup size and firmness for everyone on real photos. The partial mask separates the two parts of the loss: with age supervised the children's gain stays, so the age label carries it. Gender falls to chance although its label stays supervised. A reading that was not tested: Final's gender gain depends on the seven constants on the 3DPW men as much as on the gender label.

Before the first masked run I fixed a rule: a masked model would replace Final if gender accuracy stayed at 60 % or more, the AUC at 0.7 or more, and the spread of muscle and weight rose above 0.08. Both masked runs fail on gender while passing on the AUC, so Final stays the final model and the Student's teacher.

Not shown: whether the varied values are right (no real data set I have gives muscle or weight), whether Final-partial-mask brings the variety back (it had no demo run), and why gender falls.

6.9. Baseline comparison and model cost

Table 6.4 sets NAVER's released Anny model, run with our code, labels and settings, next to ours. On the AnnyOne holdout it reads 60.3 mm PVE against 73.7 for Final. NAVER's model was trained on the AnnyOne release, which includes the 100 images we hold out, so its 60.3 mm is most likely an in-sample number; ours are held out. On 3DPW, which neither model saw before our

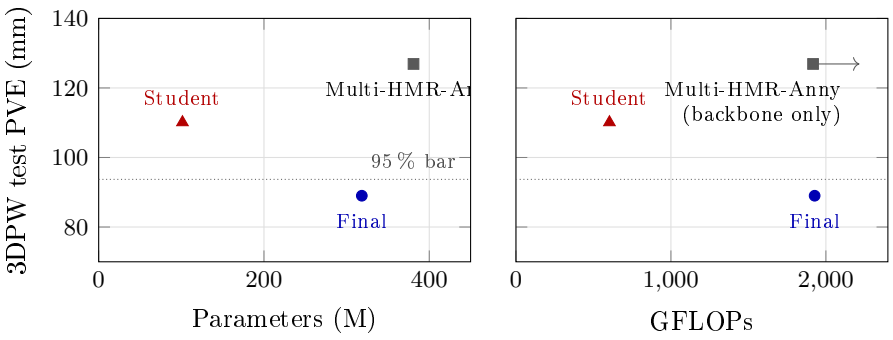


Figure 6.10.: *Model size and compute against 3DPW-test PVE (corrected labels, all 8,190 images; one run per model). Multi-HMR-Anny's GFLOPs count its backbone only (arrow: the true value is higher). Dotted line: the 95 % bar against Final (93.7 mm).*

fine-tune, the final model is 37.9 mm better (89.0 vs 126.9). Both networks use the same ViT-L backbone; the difference is the head, NAVER's 8-layer cross-attention decoder with 76.57 M parameters against 14.01 M in ours. Figure 6.10 sets size and compute against the 3DPW error, the Student included.

The fair comparison is 3DPW, but most of Final's lead there comes from fine-tuning on 3DPW, which NAVER's model never had: Control, without our fixes, is already at 91.5 mm. The comparison shows what the real-image fine-tune buys, not a better network. Final is about one sixth smaller than NAVER's model; that is not the "significantly smaller" model of requirement 1b, which is why the Student exists.

Not shown: NAVER's model fine-tuned on our labels, which would separate the network from the data, and its latency and total FLOPs; its published SMPL-X numbers are not comparable.

6.10. The ViT-B student

The Student follows the recipe of Table 4.3: a ViT-B backbone started from NAVER's Multi-HMR ViT-B and heads copied from Final, distilled from Final for 30 epochs. The cost side of Table 6.5 is met. The accuracy side is not: on the AnnyOne holdout the Student reaches 119.6 mm PVE against the teacher's 73.7 (bar 77.6), 1.62 times the teacher's error, and on the 3DPW test 110.1 mm against 89.0 (bar 93.7), 1.24 times. MPJPE and PA-PVE miss too; recall stays within the rule. Figure 6.11 shows that the holdout error was still falling at the last epoch, from 139.5 mm at epoch 15 to 122.8 mm at epoch 30, while the 3DPW validation error had flattened (88.0 to 87.7 mm in the last five epochs). Against Multi-HMR-Anny, the Student is 16.8 mm better on 3DPW (110.1 vs 126.9), which meets NAVER's 3DPW bar of 133.6 mm; it misses NAVER's AnnyOne bar of 63.5 mm, which rests on a most likely in-sample number.

By the 95 % rule against Final, the Student fails on both sets: its error is 62 % above the teacher's on AnnyOne and 24 % on 3DPW, where the rule allows about 5 %. The curves suggest that a longer run would narrow the

Metric	Teacher	Student	Ratio	Bar	Met?
AnnyOne holdout, V1					
PVE (mm)	73.7	119.6	1.62	$\leq$ 77.6	no
PA-PVE (mm)	50.3	80.8	1.61	$\leq$ 52.9	no
MPJPE (mm)	77.9	127.2	1.63	$\leq$ 82.0	no
Recall (%)	87.0	85.4	0.98	$\geq$ 82.7	yes
AnnyOne test-100, V0					
PVE (mm)	77.5	122.7	1.58	$\leq$ 81.6	no
PA-PVE (mm)	54.4	84.2	1.55	$\leq$ 57.3	no
Recall (%)	87	85	0.98	$\geq$ 82.7	yes
3DPW test (v23 labels), V0					
PVE (mm)	89.0	110.1	1.24	$\leq$ 93.7	no
PA-PVE (mm)	57.6	71.8	1.25	$\leq$ 60.6	no
MPJPE (mm)	88.3	112.0	1.27	$\leq$ 92.9	no
Recall (%)	99	99	1.00	$\geq$ 94.1	yes
Cost (L40S, batch 1, one person)					
Params (M)	318.38	101.38	3.14×	fewer	yes
GFLOPs	1927.4	603.3	3.19×	fewer	yes
ms per image	59.7	28.5	2.10×	faster	yes

Table 6.5.: *The Student against its teacher (Final) and the 95 % rule: every error of the Student at most the teacher's divided by 0.95, every recall at least 0.95 of the teacher's, and fewer parameters, fewer FLOPs and a faster forward pass. The rule fails on every error and holds on recall and cost. One run, no seed. Against Multi-HMR-Anny the PVE bars would be 63.5 mm on AnnyOne (most likely in-sample for it; not met) and 133.6 mm on 3DPW (met). Cost: both models timed in the Student's job on one L40S.*

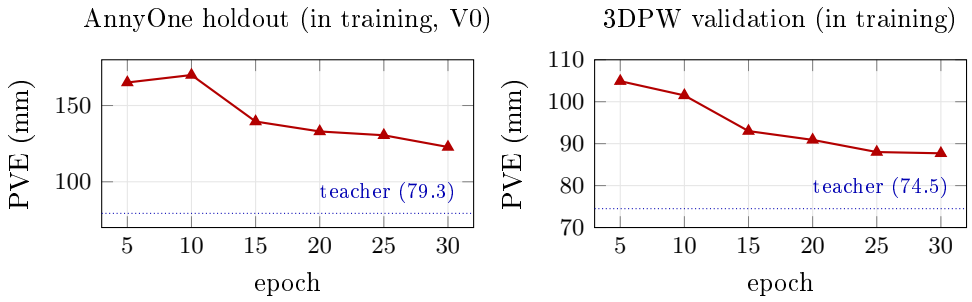


Figure 6.11.: *The Student's PVE during training, every five epochs: left the AnnyOne holdout as evaluated during training (setting V0), right the corrected 3DPW validation labels. Dotted: the teacher's value at the last epoch of its own training (Figure D.6). The heads were frozen in the first five epochs. One run, no seed; values from the training log.*

AnnyOne gap, where the error still fell by about 8 mm per five epochs, but not the 3DPW gap, where it had flattened. What holds is the comparison with Multi-HMR-Anny: a network with 3.76× fewer parameters is better on real images, although most of that lead likely comes, as for Final, from fine-tuning on 3DPW. Which reference sets the bar is the supervisor's decision.

Not shown: a second run, a longer schedule, and the Student's shape, fingers and necks on real photos.

6.11. History: the ViT-S students and the schedule to Base

Run	Change against the previous run	PVE	PA-PVE	MPJPE	Rec.
L-stage 1	fresh heads, all blocks trained (side branch)	–	–	–	–
L-stage 2	fresh heads, backbone frozen	145	–	–	–
L-stage 3	+ last 4 of 24 blocks, helper mask	73.6	–	–	–
L-stage 4	+ neck and hand loss ×4	73.6	56.5	73.2	–
Base	+ shape loss weight 20	74.3	57.5	74.4	90

(a) *The transfer schedule that produced Base (Table 4.2); recall (Rec.) in %. L-stages 2–4: evaluation during training; Base: the standalone script.*

			AnnyOne holdout, V0					3DPW test, v23[c]	
Model	Teacher	Params	PVE	PA-PVE	MPJPE	Prec.	Rec.	PVE	PA-PVE
Base (ViT-L)	–	318.4	74.3	57.5	74.4	81	90	121.8	72.0
Mixed-L	–	318.4	78.5	57.4	78.6	83	89	95.5	62.9
Student-S/2	Base	32.8	193.4	128.2	210.7	–	86	–	–
Student-S/6b[a]	Base	37	126.9	92.1	132.0	77	88	–	–
Student-S/6c	Base	37	**114.5**	**84.6**	**118.4**	81	89	142.6	91.0
Student-S/mixed[b]	Base	37	120.6	88.2	125.2	78	87	–	–
Student-S/final[b]	Mixed-L	37	120.6	86.9	125.0	76	87	121.6	83.7

(b) *The ViT-S students against their ViT-L teachers; parameters in millions, precision (Prec.) and recall (Rec.) in %. Student-S/2: feature distillation only; Student-S/6b: heads fed through a projector at the teacher's width; Student-S/6c: Student-S/6b at batch 16; Student-S/mixed and Student-S/final: continued with 3DPW images. [a]Evaluation during training. [b]30 % 3DPW images per batch, old labels. [c]Corrected labels, the script of Table 6.1b; Final: 89.0 / 57.6 mm.*

Table 6.6.: *History, before the depth fix, the bone correction and the corrected 3DPW labels. AnnyOne holdout (100 images), setting V0, errors in mm; one run per model, no seed. "–": not measured with the same script. The students' old-label 3DPW numbers are retired.*

Table 6.6 keeps the draft's V0 measurements from before both fixes: the schedule to Base (Table 6.6a) and the ViT-S students (Table 6.6b), at best 1.54 times their teacher's error. Three lessons followed: a backbone trained for mesh recovery gave the largest gain, so the Student starts from NAVER's ViT-B; heads at the teacher's width beat narrower ones, so it keeps Final's heads; and data limited the students, so it sees six times more images.

7. Discussion

This chapter reads the results against the requirements, explains why the headline error barely moved, states the limits of each claim and records what the work taught about building such a pipeline.

7.1. The requirements, one by one

The supervisor set three tiers (Section 1.3): requirements without which he would not consider the thesis ready for submission (1a, 1b, 1c), requirements for an acceptable grade (2a, 2b), and a well-written report with verifiable code for the best grade (3). Table 7.1 gives the status of each on 19 October 2026 and its evidence. I split 1a into pose and shape, because their evidence differs. The paragraphs below name only what is missing and where Chapter 8 plans it.

1a-pose (partly). Missing are correct near and far depth (slope 0.42; Figure 6.2), a pose number on real photos, where placement offsets, ghost bodies and hard poses such as the seated guitar player remain, and a second run. Section 8.2 plans a repeat of Final; the slope stays an open limitation.

1a-shape (partly). Gender and age change with the person; muscle, weight, cup size and firmness do not, because Final learned the constants the converter writes for every 3DPW person, and neither mask arm freed them without losing gender (Section 6.8). Missing are a measure of how well each phenotype follows its label on AnnyOne, which Section 8.2 plans, and a recipe for that variety.

1b (not met). Final is Multi-HMR with its output resized to Anny, and only 16 % smaller than NAVER's model. The Student is 3.14× smaller than Final but misses the 95 % bar on both sets (Section 6.10); read against Multi-HMR-Anny instead, it meets the 3DPW bar and misses the AnnyOne bar, which rests on a likely in-sample number. Missing are the supervisor's choice of the reference model and his view on whether a Multi-HMR-style network with a smaller backbone counts; Section 8.2 plans the next step for each answer.

1c (met with caveats). The caveats are those of Section 3.6: the labels test pose, gender, height and proportions but not body build, and no number compares with published 3DPW results. A body-joint MPJPE next to the 163-joint one would help, but is not written.

Req.	Supervisor's wording (short)	Status	Evidence	What is missing
1a-pose	"Pose of all body parts must look correct" on (a) the 3DPW test set and (b) any in-the-wild image	**partly**	3DPW test: PVE 89.0 mm, recall 99 % (Table 6.1b); neck and finger flips corrected on the holdout (Sections 6.3, 6.4); real photos by eye (Figures 6.6, 6.4)	depth slope 0.42 (Figure 6.2); a second run
1a-shape	"Mesh size, proportions, age, muscles, and other aspects of the 3D shape must visibly change according to the person"	**partly**	Relative Human: gender 64.2 % (Base 46.5), child AUC 0.719 (Base 0.606) (Table 6.2); muscle, weight, cup size, firmness nearly constant on photos; both mask arms lose gender (Table 6.3)	per-phenotype agreement on AnnyOne; variety without losing gender
1b	"Must not be exactly the same as the Multi-HMR model": new and better on both test sets, or "significantly smaller" and ≥ 95 % as good	**not met**	Final is Multi-HMR with an Anny head, -16 % parameters against Multi-HMR-Anny (Table 6.4); Student 3.14× smaller, 119.6 / 110.1 mm, misses the bars 77.6 / 93.7; beats Multi-HMR-Anny on 3DPW (Table 6.5)	the supervisor's choice of the reference model and whether a smaller student counts
1c	Evaluation on the real-world set with Anny parameters, not only SMPL-X	**met with caveats**	3DPW test in Anny space, 8,190 images; label check WARN (old labels FAIL) (Table 3.4)	nothing required; caveats of Section 3.6; a body-joint MPJPE
2a	Convert 3DPW train and test to Anny and include the train set in training	**met**	train 5,882 images / 7,584 people, 30 % of each batch (Section 4.6); 3DPW test: Control 91.5 against Base 121.8 mm (Table 6.1b)	nothing (a larger real-image share is optional)
2b	The model understands motion in video instead of predicting every frame on its own	**not started**	no temporal code; every model runs frame by frame	a temporal model and its evaluation (Section 8.2)
3	A well-written report that shows understanding; a clean codebase and trained models he can verify	**partly**	this thesis; Appendix A maps every number to its run, Appendix C lists every setting	released checkpoints, clean paths, licences checked (Section 8.2)

Table 7.1.: *The supervisor's requirements, in his order, and their status on 19 October 2026. AnnyOne holdout in setting V1; 3DPW test on the corrected (v23) labels, setting V0; Relative Human validation split. One run per model, no seed. Section 7.1 discusses each row.*

2a (met). His expectation held: the real images, not the fixes, brought the large 3DPW gain. Nothing is missing.

2b (not started). No temporal code exists. Section 8.2 plans a small refiner on the frozen Final model; whether a refiner on frozen features counts as the temporal model he means is a question for him.

3 (partly). No checkpoint is released, the scripts still carry the cluster's paths, and the licences have not been checked (Section 7.3). Section 8.2 lists the release steps.

7.2. Why PVE barely moved while everything else did

From Control to Final the root error fell by 48 %, neck02 from 175.8 to 4.8 deg and the finger error from 17.4 to 8.3 deg, while PVE fell only from 77.4 to 73.7 mm, within run-to-run variation, and Base keeps the lowest PVE (Table 6.1a). Three facts explain most of this.

PVE cannot see placement. PVE aligns the pelvises before it measures, so root error, depth and drawn size are removed by construction; on Base the correlation between a person's PVE and their depth error was -0.03 (Table 5.2).

The bone flips cancelled. The half-turns of neck02 and the head bone cancel, so Base's head pointed the right way and only the neck skin between them was twisted: 171 of 13,718 vertices. Its error fell from 27.7 to 11.0 mm, about 0.2 mm spread over the whole mesh (an estimate, not a measurement).

Noise floor and restart cost. One run resolves about 10 mm PVE (Section 5.7), so the gap of 3.7 mm supports neither a gain nor a cost. Restarting a fine-tune of Base already cost about 2.4 mm, and Control sits 6.9 mm above Base, likely because every fine-tune moves away from the synthetic domain of the holdout; this was not tested.

What did move is PA-PVE, which also removes global rotation and scale: Final is 6.1 mm below Control and 3.0 mm below Base, and the pilots tie part of this to the bone correction (Table 5.4). The likely reason is that the error in global rotation dominates PVE and hides the articulated part where the flips sit; a per-region breakdown would test this. I therefore report each fix with the measure that can see it, and keep PVE in every table as the check that a fix costs nothing a single run can resolve.

7.3. Limitations

Children: a measured improvement, not solved. Only 49.3 % of the babies and 16.8 % of the kids fall below the child line, and the boy of the demo photos gets an adult body. Children are rare in AnnyOne and absent from the holdout,

and the 3DPW labels pin every real person's age at 0.55. Whether the domain gap or an L1 loss through a sigmoid on a rare value dominates, the evidence cannot rank.

Depth: the slope did not move. The fix moved the offset and the size, not the slope (0.42): 94.4 % of all people, the near ones most, are still placed too far. The cause is unknown, and all depth numbers are synthetic.

One run per arm, no seed. The claims rest on gaps far above the run-to-run variation (root error, bone angles, girth, fingers, PA-PVE); the PVE gap, the 3DPW gain of the fixes and the recall cost lie within or at its edge. There are no bootstrap intervals, and the holdout holds 100 images with 222 people. The finger gain, likewise, is measured on the synthetic holdout only; on real images it rests on three images by eye.

The recall cost. Final finds 3.2 points fewer people than Base, beyond the variation, and every depth-fix pilot lost recall against its control. I state it as a real cost; the head-anchored detection cell is the likely reason, not yet tested.

What the 3DPW labels can teach and test. The labels carry a neutral body, with 8 of 11 phenotypes constant and the neck, head and hand bones at rest (Section 3.6). They cannot supervise body build or fingers or test the neck correction, they put a floor of about 14 mm under every number, and precision is a lower bound because bystanders count as false positives.

The comparison with NAVER's model. Its AnnyOne number is most likely in-sample (Section 6.9). On 3DPW, Final was fine-tuned on 3DPW-train and NAVER's model was not; NAVER's model fine-tuned on our labels would separate network from data and was not run.

Licences. SMPL and 3DPW are licensed for non-commercial research, Relative Human ships without a licence file (I used only its validation split, for evaluation), and AnnyOne and NAVER's checkpoints carry their own terms. I have not yet checked whether these allow releasing derived weights.

7.4. Engineering lessons

The costly faults were silent: each gave plausible output and showed only when one stage was measured on its own. Four habits found them. I measured each stage alone; only an independent check of the finished label files found the three label defects (Section 3.5). I read every change against a control, which kept the neck pilot's PVE and the 3DPW gain from being credited to the fixes. I checked by geometry, not by names: the reversed spine, the reversed gender value, the pelvis targets and the half-turned neck bones all hid behind plausible names or group means. And every change entered the code as a switch whose default keeps the old behaviour (Appendix C).

8. Conclusion and Future Work

8.1. Conclusion

I moved Multi-HMR to the Anny body model and trained it on synthetic AnnyOne images and on real 3DPW images that I converted to Anny. The model I started from had three visible failures. I first tested what was not their cause and then traced each one to a mechanism: training targets built at the pelvis while the network anchors each person at the head, and four bones turned by half a turn at the output. The fixes are small, and their effect is large where it can be measured. Against a control with the same training, Final cut the root error from 1,402 to 728 mm, the neck02 error from 175.8 to 4.8 degrees and the finger error from 17.4 to 8.3 degrees, while the holdout PVE (73.7 against 77.4 mm) stayed within run-to-run variation. The main lesson is about measurement: the standard error measure could not see the worst defect, and each failure became tractable only once a number for it existed.

In the order of the supervisor's requirements: for the pose (1a), Final has 89.0 mm PVE on the 3DPW test set against our labels, with necks and fingers corrected on the holdout; on real photos the pose is judged by eye, and predicted fingers, shown in Figure 6.4, look like hands in Final and are splayed and stiff in Base on the images where the hands are large enough to judge. For the shape (1a), Final recognises gender in 64.2 % of the Relative Human people (Base 46.5 %) and separates children from adults with an AUC of 0.719 (Base 0.606); children are a measured improvement, not solved, and muscle and weight barely vary between people on real photos. For 1b, the Student has 3.14× fewer parameters and 3.19× fewer FLOPs than Final, but its error is 1.62 times the teacher's on AnnyOne and 1.24 times on 3DPW, so 1b is not met against Final; on 3DPW it is still better than NAVER's Multi-HMR-Anny (110.1 against 126.9 mm). For 1c and 2a, the converted 3DPW labels exist with three defects fixed and an independent check, and training on them brought the 3DPW error from 121.8 (Base) to 91.5 mm (Control) before any fix. Motion in video (2b) was not started. For 3, every number traces to a run, a log and a checkpoint (Appendix A); the code is not yet cleaned up or released. Every model was trained once without a fixed seed, and the 3DPW numbers are not comparable with published 3DPW results; Section 7.3 lists the limitations.

8.2. Future work

The steps below are planned, each with its state on 19 October 2026.

The reference for 1b. The next step is the supervisor's decision on which model sets the 95 % bar. If it is Final, the planned option is to train the Student longer from its last checkpoint, since its AnnyOne error was still falling; this needs a resume mode that is not yet written, and the ratio will be reported as measured. A ViT-L cut to its first 12 blocks would be only about two times smaller and is not planned. If it is Multi-HMR-Anny, the Student meets the 3DPW bar, and the AnnyOne bar rests on a likely in-sample number; I would then report both ratios with that caveat.

Evidence that shape follows the person (1a). The rest-pose line-up (Figure 6.9) is a by-eye reading. Still planned is a number: per-phenotype correlation, slope and R^2 between prediction and label on the AnnyOne holdout, from per-person files that already exist.

A measured spread for Final. A second run of Final with the same recipe would replace "within run-to-run variation" by a measured spread. The training code sets no seed, so this is a repeat, not a seeded run. It takes about two GPU hours and is planned for the fourth week if a slot is free.

Motion in video (2b). In the second month I plan a small temporal refiner on the frozen per-frame model. It reads the per-person tokens of Final over windows of 16 frames (1.6 s at the 10 frames per second of the labels) with a two-layer transformer and pools the body shape over each track. It is trained on the 3DPW training tracks and judged by acceleration error and PVE against the per-frame model, shape averaging and temporal smoothing. If it does not beat smoothing, the thesis will say so.

Code and model release (3). The code will get one script that reproduces the evaluations, paths set by environment variables, an environment file and a rewritten README. Final, Final-masked, Control, Base and the Student will be released with checksums, subject to the licence terms of AnnyOne, NAVER's checkpoints and 3DPW.

Open beyond this plan. Two problems have no plan inside this thesis. Children still receive adult-shaped bodies too often; weak age labels from the Relative Human training split or child-rich synthetic data are candidates. The depth slope of 0.42 means that near and far people still get similar depths; the likely reason, a pull toward the typical training distance, was not tested.

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A. Reproduction: Model Names, Runs and Job Ids

Every number in this thesis can be traced in two steps. Each headline number is a macro in one of the files `numbers_*.tex` of the thesis source, and the comment next to it names the Slurm job and the line of its log, for example `3611336:191`. Table A.2 maps that job to the model, the log and the checkpoint, and Table A.1 maps every model name to the run name it had during the project. This appendix is the only place where job ids and old run names appear.

A.1. Model names

"L" is our ViT-L line, every model that keeps the ViT-L backbone. L-stage 1 to 4 are the steps of the transfer schedule that turned NAVER's released model into ours; Base is its end and the start of every later fine-tune. Final and Control are the two arms of the last experiment, the same 10-epoch fine-tune of Base with and without the two fixes. The number after "Student-S/" is the old version number of a ViT-S student; Student is the ViT-B model distilled from Final.

Table A.1.: *Model names: the thesis name of every model, the run or checkpoint name it had during the project ("ep." = the epoch kept), the weights it started from, the backbone, the training data and the schedule, with the setting that is new in the row. Backbone: L, B, S = DINOv2 ViT-L, ViT-B, ViT-S with 14-pixel patches; "all" = every block trained, "last 4" = the last 4 of 24 blocks, "+p" = a linear projector to the heads' width 1024. Data: A = AnnyOne; A+P = 70 % AnnyOne and 30 % 3DPW train, with the old labels (before their correction, Chapter 3) or the corrected v23 labels. Schedule: epochs × iterations per epoch × batch, then the learning rate, halved every 30 epochs unless noted (planned epochs where the run stopped early). Checkpoints lie in `logs/anny_model/<run>/checkpoints/` with copies in `models/multiHMR/`; Table A.2 gives the path of every model with a number in the thesis, and Appendix C the full flags of Final, Control and the Student. "Not recorded": no script, log or document in the repository states it.*

Thesis name	Old run / checkpoint name	Starting weights	Backb.	Data	Schedule; what is new
Released starting point and baseline					

Thesis name	Old run / checkpoint name	Starting weights	Backb.	Data	Schedule; what is new
Multi-HMR-Anny	`multiHMR_672_L_anny` (NAVER's release)	trained by NAVER	L, own decoder	A, BEDLAM, COCO, MPII	not recorded; evaluated with its own model class
Our ViT-L line ("L"), the transfer schedule					
L-stage 1	`anny_s1_pretrained_bb_v2`	Multi-HMR-Anny backbone, fresh heads	L, all	A	$150 \times 1{,}000 \times 4$, 10^{-4}, from epoch 177 at $3 \cdot 10^{-5}$; side branch, no checkpoint kept
L-stage 2	`anny_s2_frozen_bb_v2`, ep. 299 ("v2")	Multi-HMR-Anny backbone, fresh heads	L, frozen	A	$150 \times 1{,}000 \times 4$, 10^{-4}, from epoch 283 at $3 \cdot 10^{-5}$
L-stage 3	`anny_s2_partialft_v3`, ep. 54 ("v3")	L-stage 2, epoch 299	L, last 4	A	$300 \times 1{,}000 \times 4$, 10^{-5}; helper bones held at rest
L-stage 4	`anny_s2_boosted_v4`, ep. 299 ("v4")	L-stage 3, epoch 54	L, last 4	A	$300 \times 1{,}000 \times 4$, 10^{-5}; neck and hand bones weighted $\times 4$
Base	`anny_s2_shape_v5`, ep. 99 ("v5")	L-stage 4, epoch 299	L, last 4	A	$100 \times 1{,}000 \times 4$, 10^{-5}; shape weight 20
Mixed-L	`anny_teacher_v9_mixed`, ep. 62 and 149 ("v9")	Base	L, last 4	A+P, old	$150 \times 1{,}000 \times 4$, 10^{-5}, halved every 60
Fresh-heads-L	`anny_v10_scratch_mixed`, ep. 290 ("v10", not from scratch)	Multi-HMR-Anny backbone, fresh heads	L, last 4	A+P, old	$300 \times 1{,}000 \times 4$, 10^{-5}; 3DPW from the first step, no neck or hand weight
The last experiment: two arms and two shape-mask variants					
Control	exp7 `v5_root_dist1_nofix_v23_e10`, ep. 9; job 3611339	Base	L, last 4	A+P, v23	$10 \times 1{,}000 \times 4$, 10^{-5}; 3DPW hands masked, pelvis targets
Final	exp7 `v5_pc_dist5_bonefix_v23_e10`, ep. 9; job 3611336	Base and its bone correction	L, last 4	A+P, v23	as Control, with head targets, depth and translation weights 5, bone correction
Final-masked	exp7 `v5_pc_dist5_bonefix_v23_e10_shpmask`, ep. 9; job 3612325	as Final	L, last 4	A+P, v23	as Final, 8 converter constants masked for 3DPW people
Final-partial-mask	exp7 `v5_pc_dist5_bonefix_v23_e10_shpmask_noage`, ep. 9; job 3614042	as Final	L, last 4	A+P, v23	as Final, the 7 constants other than age masked; not adopted
ViT-S students ("Student-S"), history; teacher Base unless noted					
Student-S/1	distillation v1, `anny_distill_vits`	ImageNet DINOv2-S, random heads	S, all	A	$300 \times 1{,}000 \times 4$, 10^{-4}; feature distillation
Student-S/2	v2, `anny_distill_vits_v2_a`	Student-S/1, epoch 199	S, all	A	$200 \times 1{,}000 \times 4$, $2 \cdot 10^{-5}$, halved every 60
Student-S/3	v3, `anny_distill_vits_v3`	ImageNet DINOv2-S; 54 of 67 head tensors from Base	S, all	A	$150 \times 1{,}000 \times 4$, $5 \cdot 10^{-5}$, halved every 60; output distillation
Student-S/4	v4, `anny_distill_vits_v4`	NAVER's `multiHMR_672_S` backbone; heads as /3	S, all	A	$150 \times 1{,}000 \times 8$, $7 \cdot 10^{-5}$, halved every 60

Thesis name	Old run / checkpoint name	Starting weights	Backb.	Data	Schedule; what is new
Student-S/5	v5, `anny_distill_vits_v5`	as Student-S/4	S, all	A	as /4; heads frozen for 5 epochs
Student-S/6	v6, `anny_distill_vits_v6`, ep. 137	NAVER's ViT-S backbone; all 67 head tensors from Base	S+p	A	as /4; stopped at epoch 138 by the time limit
Student-S/6b	v6b, `anny_distill_vits_v6b`, ep. 140	Student-S/6, epoch 137	S+p	A	150 × 1,000 × 8, $2 \cdot 10^{-5}$, halved every 60; teacher in bfloat16
Student-S/6c	v6c, `anny_distill_vits_v6c`, ep. 149	Student-S/6b, epoch 140	S+p	A	150 × 1,000 × 16, $3 \cdot 10^{-5}$, halved every 60
Student-S/mixed	v7, `anny_distill_vits_v7_mixed`, ep. 130	Student-S/6b, epoch 140	S+p	A+P, old	150 × 1,000 × 8, $2 \cdot 10^{-5}$, halved every 60
Student-S/final	v8, `anny_distill_vits_v8_final`, ep. 52	Student-S/6c, epoch 149; teacher Mixed-L	S+p	A+P, old	150 × 1,000 × 16, $3 \cdot 10^{-5}$, halved every 60
ViT-B student (requirement 1b)					
Student	exp8 `student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e30`; job 3614191 (done 19 Oct)	NAVER's `multiHMR_672_B` backbone; every head tensor from Final; teacher Final	B+p, all	A+P, v23	30 × 1,000 × 8, $3 \cdot 10^{-5}$, halved every 12; Final's targets and masks; holdout 119.6, 3DPW 110.1 mm, 95 % rule not met

A.2. Environment

All training, evaluation and demo jobs ran on the DFKI Pegasus Slurm cluster, from the repository clone in `/netscratch/najib/multi-hmr`. The repository is public on GitHub (`saadnajib/Multi-Person-Whole-Body-Human-Mesh-Recovery-in-a-Single-Shot-with-Annyone-dataset-and-3dpw-dataset`); checkpoints, logs and datasets are not in it. Each job script prints the short git hash of the clone into the run's `settings.txt`, so a run can be matched to a commit.

Every job runs inside the container image `ubuntu+20.04_v4.sqsh` (in `/netscratch/najib/ContainerImages/`), started by `srun` with the option `--container-image` and the mounts `/fscratch`, `/netscratch`, `/home/najib` and `/dev/shm`. Inside it, `slurm/env.sh` is sourced first. It activates the conda environment `multihmr` (Python 3.9, PyTorch 2.0.1 with CUDA 11.7, xformers 0.0.20, torchvision 0.15.2), moves the Hugging Face, Torch and Anny caches to `/netscratch/najib`, sets one thread per data-loader worker, and strips the `X | None` type hints from the downloaded DINOv2 hub code, which Python 3.9 rejects. `slurm/egl.sh` then installs the EGL libraries for the headless renderer; a failure there only disables the mesh overlays.

That PyTorch build has GPU kernels up to compute capability 8.6, so the training and evaluation scripts ask for A100, A40, L40S and RTX A6000 nodes and refuse an H100 or H200 (compute capability 9.0). The V100 nodes `gifu` and `glendale` are excluded: every converter task sent there

failed within a minute with an argument-parser error whose cause is unknown. Only the 3DPW converter ran on H100 nodes, through a small virtual environment with PyTorch 2.1.2 for CUDA 12.1 on top of `multihmr` (`PY_VENV=/netscratch/najib/venv_cu121`). Steps that need only NumPy (building the 3DPW label files, the re-fit check) run on the login node. The model-cost script needs a GPU node, because `model.py` imports the renderer and the login node has no EGL.

A.3. Runs, jobs, logs and checkpoints

Table A.2 lists every job behind a number, a table or a figure of the thesis; all of them have finished.

Table A.2.: *The Slurm jobs behind every number, table and figure of the thesis, with the log and the checkpoint or output. Model names as in Table A.1. Paths are relative to the project folder on the cluster; the log of job j is `logs/<logname>_j.out` (array jobs: `_j_<task>.out`). E7 stands for `results/experiments/exp7_final_model/`, M for `models/multiHMR/`.*

Model or item: what the job did	Job(s)	Log name; checkpoint or output
Models		
Base: holdout trace (V1 PVE, placement)	3570222; re-trace 3571596	`diag_holdout`; `logs/anny_model/anny_s2_shape_v5/checkpoints/00099.pt`
Mixed-L: holdout trace	3571963	`diag_holdout`; M/`anny_teacher_v9_mixed_00062.pt`
Fresh-heads-L: holdout trace	3571794	`diag_holdout`; `logs/anny_model/anny_v10_scratch_mixed/checkpoints/00290.pt`
Final: training, holdout trace, 3DPW test	3611336 (smoke test 3611333)	`exp7_final_model`; E7/`v5_pc_dist5_bonefix_v23_e10/train/checkpoints/00009.pt`, copy M/`exp7_v5_final_00009.pt`
Control: the same	3611339	`exp7_final_model`; E7/`v5_root_dist1_nofix_v23_e10/`
Final-masked: the same	3612325	`exp7_final_model`; E7/`v5_pc_dist5_bonefix_v23_e10_shpmask/`
Final-partial-mask: the same (done 19 Oct)	3614042	`exp7_final_model` (holdout table lines 191–215); E7/`v5_pc_dist5_bonefix_v23_e10_shpmask_noage/`
Student: distillation from Final, 30 epochs, own evaluation (done 19 Oct)	3614191 (smoke test 3613926)	`exp8_distill_vitb`; `results/experiments/exp8_distill_vitb/student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e30/train/checkpoints/00029.pt`; `final_table.json` in the same run folder
Multi-HMR-Anny: holdout and 3DPW test with our code	3614192 (3613928 crashed: FFN width)	`eval-test3dpw` (task 0); M/`multiHMR_672_L_anny.pt`
Model cost: parameters, GFLOPs, latency of Multi-HMR-Anny and Final (A100)	3614947	`model_cost`; `results/diagnostics/model_cost_naver_vs_final.json`
Evaluations and demos		
3DPW test (v23): Base, Mixed-L, neck pilot (Final and Control in their own jobs)	3612073, 3612074, 3612075	`eval-test3dpw` (task 0), lines 117–125

Model or item: what the job did	Job(s)	Log name; checkpoint or output
3DPW test (v23): Student-S/6c, Student-S/final, also AnnyOne test-100	3615020_2, 3615020_4	`eval-test3dpw`, line 119 (_2) and 118 (_4)
Relative Human: Base, neck pilot, Final, Final-masked, Final-partial-mask	3611285_0, _1, 3612131, 3613616, 3616821	`eval_rh`; `results/diagnostics/relative_human_val_<tag>/`
Demos: neck pilot, Final, Final-masked, each next to Base	3581312, 3612078, 3613617	`demo-compare`; `demo_montage_photos/`, `demo_montage_3dpwtest/`
Demos: predicted fingers (Base, Final); rest pose (Base, Final, Final-masked)	3615018, 3615019	`demo-compare`; scripts `slurm/demo/run_demo_compare_hands.sh`, `_rest.sh` (copies made on the cluster)
3DPW labels		
Label check before and after the re-fit	3575369, 3611284 (v23)	`check_3dpw_conversion`; `results/diagnostics/check_3dpw_conversion_v23/`
Re-fit with converter v2.3, 87 person-sequences	3590385, 3602813	`smpl2anny-all`; `threedpw_anny_v23/labels/`; `logs/check_refit_all_v23.log`
Label rebuilds from the old fits	3579266, 3581320	`3dpw-labels`; `threedpw_anny_full/labels_rebuilt/`
Pilots and diagnostics (outputs in `results/experiments/` and `results/diagnostics/`)		
Phenotypes; youngest 50 training people	3569948; 3569950	`diag_pheno`; `diag_young`
Real-photo ages with the crop camera: Base, sampler pilot, its control	3569949, 3572419, 3572420	`diag_cropK`
Depth pilots, 2 epochs (×1, control, ×5, ×20) and 10 epochs (×1, control, ×5)	3571598, 3571599, 3572418, 3572483; 3571933, 3571934, 3572482	`exp1_depth_anchor`
Sampler pilot and control; batch ages only	3571955, 3571956; 3571601	`exp2_children`
Head-loss pilot; two identical controls (noise)	3571957; 3571958, 3571960	`exp3_neck_global_rot`; `exp4_fingers` (3571960)
Finger pilot (control 3571960: crop trace, hand size)	3571959	`exp4_fingers`
Geodesic pilot and its L1 control	3573520, 3573521	`exp5_neck_fix`
Per-bone dumps of eight checkpoints	3573518, 3573519	`diag_neck_bones`; `results/diagnostics/neck_bones/`
Bone correction: Base (with the neck pilot), Fresh-heads-L	3579014, 3579015	`exp6_bone_fix`; `results/experiments/exp6_bone_fix/anny_s2_shape_v5_00099_n200_s0/bone_fix.json`; neck pilot `M/exp6_v5_bonefix_00001.pt`

The scripts behind these jobs and the exact commands that produced the thesis numbers are listed in the file `docs/thesis/REPRODUCE.md` of the repository. Each command runs from the login node in the repository clone; the flags each training run passes are in Appendix C.

A.4. What cannot be reproduced exactly

The training code sets no random seed. The `-seed` of the holdout trace only picks which training images the bone correction is estimated on. The order in which training images are drawn (with replacement) therefore differs between

two runs of the same command, and so does the number of optimiser steps skipped for non-finite gradients (16 in Final, 10 in Control, out of 10,000). Two runs of one 2-epoch configuration differed by 7.6 mm PVE, 1.9 deg head direction and 0.8 recall points (Table 5.7); a repeat of Final or Control is expected to land within such a margin, not on the same numbers. The two arms also ran on different GPU types (Final on an A100, Control on an A40, both with TF32), which changes the speed and, at the level of rounding, the arithmetic. Evaluating a fixed checkpoint involves no sampling; that it repeats exactly across GPU types was not tested.

Three further limits apply. The history scripts of the L line and of Student-S were edited after their runs, so their recipes in Table A.1 are as recorded there, not re-run. `train.py` keeps only the last two or three checkpoints of a run, so intermediate epochs are gone. The checkpoints, the AnnyOne and 3DPW data and the Relative Human annotations are not in the repository; 3DPW and Relative Human come with their own licences.

B. SMPL–Anny Joint Correspondence

[TODO: One paragraph: how the correspondence was chosen (rest skeletons compared by geometry) and how the virtual-joint offsets were measured; point to Chapter 3.]

The correspondence used by the final converter (v2.3), with the rest-pose offset of each virtual joint. Offsets are measured on the two rest skeletons after aligning them at the hip midpoint and scaling SMPL to Anny's hip–neck length (factor 0.864); limb joints use the bone head.

SMPL joint	Anny bone	virtual-joint offset (cm)
pelvis	`root`	11.6
left / right hip	`upperleg01.L/R`	4.3
spine1	`spine03`	3–7
spine2	`spine02`	3–7
spine3	`spine01`	3–7
neck	`neck01`	0 (reference)
left / right collar	`clavicle.L/R`	5
head	`head`	—
shoulder	`upperarm01.L/R`	—
elbow	`lowerarm01.L/R`	—
wrist	`wrist.L/R`	—
hand	`finger2-1.L/R`	—
knee	`lowerleg01.L/R`	—
ankle	`foot.L/R`	—
foot	`toe1-1.L/R`	—

Table B.1.: *SMPL→Anny correspondence of the final converter. The fit rotates only the root and 18 bones; the other 144 of Anny's 163 bones stay at their rest rotation, including* `neck02`, `neck03`, `head`, *the jaw and every finger bone, so the head, hand and foot targets are reached through their parent bones. The first converter mapped* `spine1` *to* `spine01`*, 25 cm too high, and the hips to* `pelvis.L/R`*, whose heads lie 4 cm from the hip joints.*

C. Flags and Weights

This appendix lists the command-line flags of the training runs of Final and Control (Table C.1) and of the Student (Table C.2) that differ from the default in the code, or that matter although they keep it; every other flag keeps its default. The loss weights are in Table 4.1 of Chapter 4, the commands that start the runs in the file `docs/thesis/REPRODUCE.md` of the repository (Appendix A).

Table C.1.: *Training flags of Final and Control. "Value" is Final's; where Control differs, its value follows in brackets. Paths are relative to the project folder on the cluster. Both runs start from Base and see 10 epochs of 1,000 iterations at batch 4, that is 40,000 images, at a constant learning rate.*

Flag	Default	Value (Control)	Meaning
Start and network			
`--pretrained`	none	Base checkpoint	weights the run starts from, heads included
`--pretrained_remap`	1	0	no renaming of weight names; Base was saved by this code
`--backbone`, `--img_size`	ViT-S, 336	`dinov2_vitl14`, 672	DINOv2 ViT-L, 48 × 48 tokens of 14 pixels
`--freeze_backbone`, `--unfreeze_last_n_blocks`	0, 0	1, 4	backbone frozen except its last 4 of 24 blocks and final norm
`--bone_fix`	off	Base's `bone_fix.json` (`none`)	constant output rotation on neck02, head, metacarpal1.L and finger4-1.R, stored in the checkpoint (Section 4.5)
Targets and losses			
`--gt_center_joint`	root	`person_center` (`root`)	the joint the depth, translation, cell and offset targets describe: the head, which the network anchors (Control: the pelvis)
`--alpha_dist`, `--alpha_transl`	1, 1	5, 5 (1, 1)	weights of the depth and translation terms
`--alpha_shape`, `--alpha_j2d`, `--alpha_v2d`	1, 1, 1	20, 0.1, 0.1	shape and 2D loss weights
`--boost_neck_weight`, `--boost_hand_weight`	1, 1	4, 4	extra weight on neck, head and hand bones, renormalised to mean 1
`--start_2d_epoch`	10	0	2D terms from the first epoch
`--mask_helper_joints_train`	0	1	hold the 88 helper bones at identity, no rotation targets
`--use_anny_shape`	0	1	phenotype targets from the dataset
`--mask_3dpw_hands`	0	1	no rotation targets for the 38 hand bones of 3DPW people
`--mask_3dpw_shape`	0	not passed	shape mask for 3DPW people; 1 only in Final-masked and Final-partial-mask
Data			

Flag	Default	Value (Control)	Meaning
`--train_data`, `--val_anny_n`	BEDLAM, 0	AnnyOne, 100	AnnyOne for training, its last 100 images as the holdout
`--exclude_indices_file`	none	the 55 manifest images	never sample the diagnostic-manifest images
`--mix_3dpw_labels`	none	`threedpw_anny_v23/labels`	corrected 3DPW labels mixed in, 30 % of a batch on average (default ratio 0.3)
`--val_3dpw_n`	0	300	evaluate on 300 evenly spaced 3DPW validation images
Optimisation, bookkeeping, job			
`--batch_size`, `--learning_rate`	2, $5 \cdot 10^{-6}$	4, 10^{-5}	images per step; Adam learning rate
`--n_iters_per_epoch`, `--max_iter`	100, 10,000	1,000, 10,000	10 epochs of 1,000 steps
`--amp`	1	0	no mixed precision (it produced NaNs with the Anny layer); TF32 stays on
`--num_workers`, `--nb_max_ckpt`, `--log_freq`	0, 10, 100	12, 2, 50	bookkeeping; output in `results/experiments/exp7_final_model/<run>/train`
`TIME`, `EVAL_TEST` (script)		5 h, 1	Slurm limit on A100, L40S, RTX A6000; after training the holdout trace and the 3DPW test (v23)

Table C.2.: *Training flags of the Student where they differ from Final's. Every other flag, the targets, masks, loss weights and 3DPW mixing included, is as in Table C.1, so that the Student learns the corrected depth; no shape mask. 30 epochs of 1,000 iterations at batch 8, that is 240,000 images, with a 20-hour Slurm limit. The colour-jitter flags (`-brightness 0.2` and so on) are passed but read nowhere in the code, so no augmentation was used.*

Flag	Default	Value	Meaning
`--backbone`	ViT-S	`dinov2_vitb14`	DINOv2 ViT-B
`--pretrained`	none	`models/multiHMR/multiHMR_672_B.pt`	NAVER's ViT-B as the backbone start
`--pretrained_remap`, `--load_only_backbone`, `--freeze_backbone`	1, 0, 0	1, 1, 0	rename NAVER's weight names; load the backbone only and train all of it
`--head_dim`	0	1024	heads at the teacher's width, fed by a 768 to 1024 projector
`--distill_teacher_ckpt`, `--teacher_bf16`	none, 0	Final, 1	frozen teacher, its backbone in bfloat16; turns the run into distillation
`--init_heads_from_teacher`, `--freeze_heads_epochs`	0, 0	1, 5	copy every matching head tensor from the teacher; keep them frozen for 5 epochs
`--lambda_kd`, `--lambda_kd_out`	1.0, 0	1.0, 1.0	feature distillation (KL, temperature 4) and output distillation (L1)
`--bone_fix`	off	the teacher's	Final's correction in training; the Student's own, estimated on 200 training images, for evaluation
`--batch_size`, `--learning_rate`	2, $5 \cdot 10^{-6}$	8, $3 \cdot 10^{-5}$	images per step; Adam learning rate
`--max_iter`, `--lr_decay_every`	10,000, 30	30,000, 12	30 epochs; learning rate halved after epochs 12 and 24
`--eval_freq`, `--nb_max_ckpt`	1, 10	5, 3	evaluate every 5 epochs, keep 3 checkpoints

D. Extra Figures

This appendix holds the figures that the chapters refer to but do not show: the second depth example, two more bone-correction pairs, a second neck pair on 3DPW, the other demo frames and photos, all twelve demo images with predicted fingers and nine more rest-pose line-ups. Every panel is judged by eye; where a body is placed from the detector box, depth errors are hidden.



Figure D.1.: *Holdout image 579860 rendered at each model's own predicted depth: ground truth, Base, and the depth pilot (head-joint targets, depth weight ×5, 10 epochs, AnnyOne only). Before the fix the ground-truth head shows above a shrunken body placed too far away. Synthetic image; one case, not a typical one. Companion of Figure 6.1.*

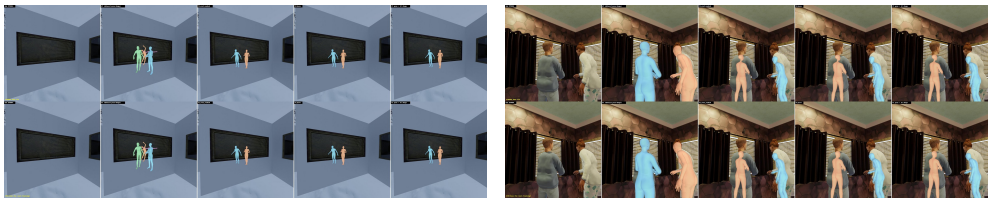


Figure D.2.: *Holdout images 579864 (left) and 579866 (right): Base as trained next to the same checkpoint with the bone correction applied at the output, no training (Section 5.4). Synthetic images at the model's own predicted depth, so the depth bug still draws the bodies too small; necks judged by eye. Companions of Figure 5.1.*

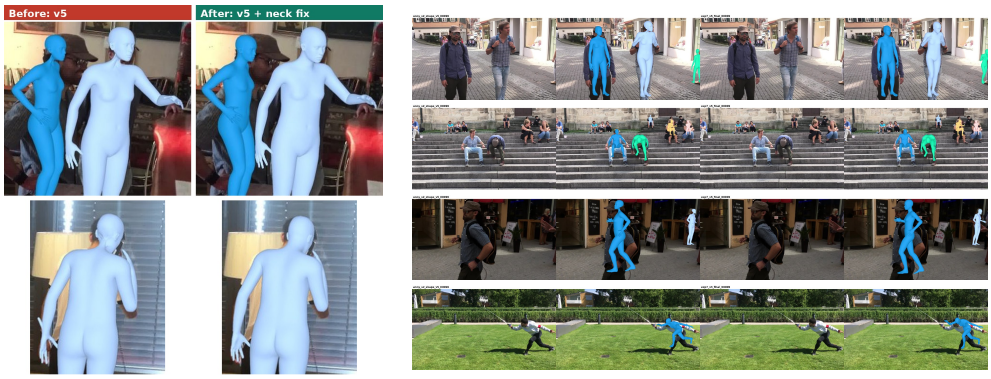


Figure D.3.: *Left: 3DPW test frames* `downtown_bar_00` *701 (top) and* `office_phoneCall_00` *440 (bottom), Base next to the neck pilot (the bone correction and a 2-epoch fine-tune); necks ringed. Bodies placed from the detector box, fingers at rest pose; judged by eye, no metric. The two columns differ by the correction and by the fine-tune.*

Figure D.4.: *Right: the other four 3DPW test frames of the demo, top to bottom* `downtown_walking`*, stairs,* `downtown_runForBus` *and fencing, each as photo and Base, then photo and Final (labels in the image). Placed from the detector box, fingers at rest pose; by eye, no metric. The first four frames are in Figure 6.6.*



Figure D.5.: *The four real photos of the demo (garden, tennis; stage, party), never used in training, each as photo and Base, then photo and Final; placed from the detector box, fingers at rest pose; by eye, no metric. The tennis photo also appears in Figure 6.8.*

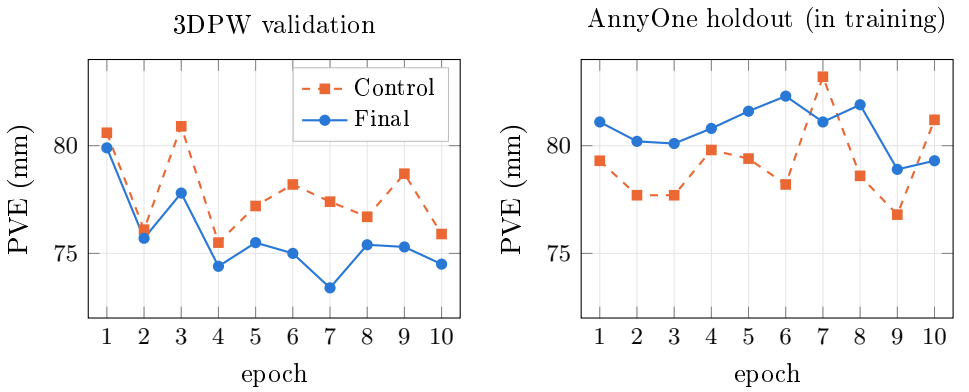


Figure D.6.: *PVE per epoch during training, Final and Control (one run each, no seed). Left: 300 images of the corrected 3DPW validation labels. Right: the AnnyOne holdout as evaluated during training, setting V0 (helper bones unmasked), so higher than the V1 numbers of Table 6.1a. Values from the training logs; read in Section 6.5.*



Figure D.7.: *All twelve demo images with predicted fingers (not set to rest pose), each as photo and Base, then photo and Final (labels in the image). Row by row: the garden and tennis photos; the stage photo and fencing; stairs and* `downtown_walking`*;* `downtown_runForBus`*, the party photo and guitar;* `downtown_bar`*,* `office_phoneCall` *and the weekly market. Demo setting, placed from the detector box; Base still has the flipped left index-finger base (metacarpal1.L) and right ring-finger base, Final has the correction. In most images the hands are a few pixels wide and cannot be judged; Figure 6.4 zooms on the three where they can. By eye; no metric.*



Figure D.8.: *The other nine demo images in rest pose, each as photo and Base, photo and Final, photo and Final-masked (labels in the image). Top to bottom: the garden photo, the 3DPW test frames fencing, stairs,* `downtown_walking` *and* `downtown_runForBus`*; then the party photo and guitar,* `office_phoneCall` *and the weekly market. Every detected person is in rest pose, so only body shape differs. Demo setting, placed from the detector box; at this size only gross differences of build can be seen. By eye; no metric. The other three images are in Figure 6.9.*

E. Pilot Results in Full

Before the final runs, six short pilots tested one change each against a control arm. Table E.1 gives, for each, what it changed, the success rule written into its script before it ran, the result and the control. All pilots fine-tune Base on AnnyOne only, at learning rate 10^{-5} and batch 4, for 2 epochs (8,000 images) unless noted, and are read on the 100-image AnnyOne holdout in setting V1. Each is one run without a seed, so every result must be read against the run-to-run variation of Section 5.7, summarised below the table. Chapter 5 explains what each result means; the full depth runs are Table 5.3.

Table E.1.: *The six pilot experiments. V1, 100-image AnnyOne holdout, matched people; fine-tunes of Base, one run each, without a seed. The rule is the success rule written into each script before the run. Head direction is the global head orientation error.*

Pilot	What changed	Success rule	Result	Control
Depth pilots (2 and 10 epochs)	training targets at the head joint instead of the pelvis; depth and translation weights ×1, ×5, ×20	slope toward 1 and intercept toward 0 m, with PVE ≤ 71.5 mm and recall ≥ 89.2 %	rule not met (PVE or recall missed in every run, controls too); root error 1.21 → 0.54 m, size 0.63 → 0.84, slope 0.33 → 0.38 (×5, 10 epochs)	same schedule, targets at the pelvis, weight ×1 (one 2-epoch, one 10-epoch)
Sampler pilot	age-balanced sampling: children (age < 0.3) per batch 2.1 → 9.7–11.1 %	10 manifest children: PVE < 47.8 mm and |age bias| < 0.078; holdout PVE ≤ 71.5, recall ≥ 89.2	inconclusive: children 48.5 vs. 53.1 mm, age bias 0.078 vs. 0.085; baby age floor 0.100 → 0.102	same run with the sampler off (age ×3 and height ×2 shape weights in both)
Head-loss pilot	L1 loss on the camera-frame orientation of head, neck03 and both wrists, weight 5	head direction < 12.0 deg; PVE ≤ 71.5 mm, recall ≥ 89.2 %	head direction 11.3 deg, within the controls' 11.7–13.6; safety limit missed (71.8 mm, 89.0 %)	two identical runs at weight 0
Finger pilot	region-normalised rotation loss (body, hands, head-neck averaged separately); a test-time person crop	fingers below the rest-pose floor 15.1 deg without a PVE cost	no effect: fingers 17.5 vs. 17.4 deg; crop 17.2 deg at body PVE 124.8 mm	same run with the loss off (one of the two identical runs above)

Pilot	What changed	Success rule	Result	Control
Geodesic pilot	geodesic angle instead of L1 on the rotation matrices	flagged bones down by ≥ 3 deg more than the control, or neck and head under 15 deg	no movement: neck02 175.6 $\rightarrow$ 175.0, head 177.9 $\rightarrow$ 177.1 deg	same run with the L1 loss
Bone-correction pilot	constant output rotation on neck02, head, metacarpal1.L, finger4-1.R; no training, then a 2-epoch fine-tune	bones < 15 deg, girth ≥ 0.9; no training: head direction $\leq$ Base + 3 deg; fine-tuned: guards against the control	no training: fails (head direction 19.1 vs. limit 15.0 deg); fine-tuned: passes (neck02 4.7 deg, girth 0.997, fingers 8.3 deg)	the geodesic pilot's L1 control

Pilot noise. Table 5.7 in Chapter 5 gives the yardstick for all of the above. Two runs of one configuration differed by 7.6 mm PVE, 1.9 deg head direction and 0.8 recall points, and restarting a fine-tune of Base cost about 2.4 mm. One pilot therefore resolves about 10 mm PVE, 3 deg and 2 recall points. The root error, the bone angles, the girth and the finger error moved far beyond that; the PVE and recall differences of every pilot did not. That is why each pilot is read against its own control, and why the two changes that entered Final were chosen for what the diagnostics showed, not for a PVE gain.

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